

Stat 6390.001: High Dimensional Probability and Statistics, Fall 2026

Topic 1

High Dimensional Probability and Statistics

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Chapter 1

Basic tail and concentration bounds

1.1 Introduction

A central challenge in high-dimensional statistics is the *curse of dimensionality*. As the dimension d grows, the sample space expands rapidly, so a fixed number of observations becomes increasingly sparse relative to the size of the space. As a result, methods and intuition that work well in low dimensions may become much less informative in high-dimensional settings.

High-dimensional problems also exhibit geometric and probabilistic behavior that can differ substantially from low-dimensional intuition. For example, if

$$X \sim N(0, I_d),$$

then the density is maximized at the origin, but a typical sample is not concentrated near the origin. Instead,

$$\|X\|_2 \approx \sqrt{d},$$

so most samples lie near a thin shell of radius approximately $\sqrt{d}$.

These phenomena motivate the study of probabilistic tools that remain useful when the dimension is large. In particular, this lecture develops tail and concentration bounds that quantify how likely a random variable is to deviate from its mean, with an emphasis on non-asymptotic guarantees.

1.1.1 One-dimensional and multivariate CLT

Let $X_1, \dots, X_n$ be i.i.d. random variables with mean μ and finite variance σ^2 . The one-dimensional CLT states that

$$\sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n X_i - \mu \right) \xrightarrow{d} N(0, \sigma^2) \quad \text{as } n \rightarrow \infty. \quad (1.1)$$

The multivariate version has the same form. Suppose now that $X_1, \dots, X_n$ are i.i.d. random vectors in $\mathbb{R}^d$ with

$$\mathbb{E}[X_i] = \mu \in \mathbb{R}^d, \quad \text{Cov}(X_i) = \Sigma.$$

Then

$$\sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n X_i - \mu \right) \xrightarrow{d} N(0, \Sigma) \quad \text{as } n \rightarrow \infty. \quad (1.2)$$

For a d -dimensional Gaussian vector $X \sim N(\mu, \Sigma)$ with positive definite covariance matrix Σ , the density is

$$f(x) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left\{ -\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right\}. \quad (1.3)$$

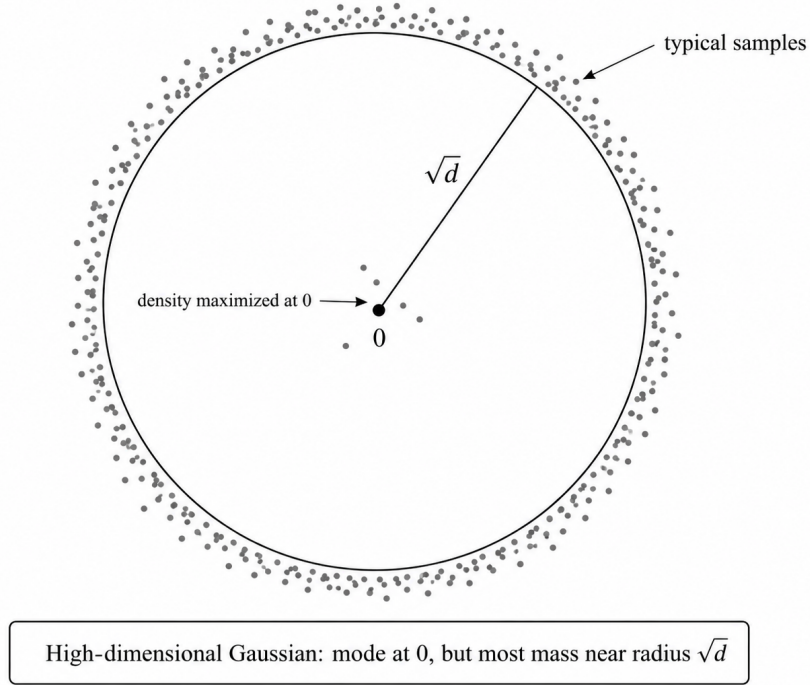


Figure 1.1: A schematic illustration of a high-dimensional standard Gaussian distribution.

Example 1.1.1 (A First High-Dimensional Phenomenon). Consider the standard Gaussian case $X \sim N(0, I_d)$. The density is maximized at the origin, so the mode is $x = 0$. However, when d is large, a typical draw is not close to the origin. Indeed,

$$\mathbb{E}\|X\|_2^2 = \mathbb{E}\left[\sum_{j=1}^d X_j^2\right] = \sum_{j=1}^d \mathbb{E}[X_j^2] = d. \quad (1.4)$$

Thus the typical Euclidean radius is of order $\sqrt{d}$; more precisely, $\|X\|_2^2$ is a chi-square random variable with d degrees of freedom and is concentrated around d . Hence most of the Gaussian probability mass lies in a thin shell with radius approximately $\sqrt{d}$, even though the density itself is largest at the origin, shown in Figure 1. This illustrates a basic feature of high-dimensional geometry: the location of the mode need not describe where a typical sample lies.

1.2 Classical Tail Bounds

The purpose of a tail bound is to control probabilities such as $\mathbb{P}(X \geq t)$ or $\mathbb{P}(|X - \mathbb{E}X| \geq t)$. The amount of information available about the distribution determines the quality of the bound. Markov's inequality uses only a first moment; Chebyshev's inequality uses a second moment; and the Chernoff method exploits the moment generating function (MGF), thereby incorporating information from all moments when the MGF exists [1].

1.2.1 Markov and Chebyshev inequalities

Proposition 1.2.1 (Markov's inequality). *If random variable X is nonnegative and $\mathbb{E}[X] < \infty$, then for*

every $t > 0$,

$$\mathbb{P}(X \geq t) \leq \frac{\mathbb{E}[X]}{t}. \quad (1.5)$$

Proposition 1.2.2 (Chebyshev's inequality). *If random variable X has finite variance and $\mu = \mathbb{E}[X]$, then for every $t > 0$,*

$$\mathbb{P}(|X - \mu| \geq t) \leq \frac{\text{Var}(X)}{t^2}. \quad (1.6)$$

Chebyshev's inequality is an immediate consequence of Markov's inequality applied to the nonnegative random variable $(X - \mu)^2$. More generally, if the k th centered absolute moment exists, then

$$\mathbb{P}(|X - \mu| \geq t) = \mathbb{P}(|X - \mu|^k \geq t^k) \leq \frac{\mathbb{E}|X - \mu|^k}{t^k}. \quad (1.7)$$

1.2.2 From Markov to the Chernoff bound

The exponential function is particularly effective because it converts sums into products and leads naturally to the moment generating function.

Lemma 1.2.3 (Chernoff bound). *Let $\mu = \mathbb{E}[X]$, and suppose*

$$\phi(\lambda) = \mathbb{E}[e^{\lambda(X - \mu)}]$$

exists for $\lambda \in [0, b]$. Then for every $t > 0$,

$$\log \mathbb{P}(X - \mu \geq t) \leq \inf_{\lambda \in [0, b]} \{\log \phi(\lambda) - \lambda t\}. \quad (1.8)$$

Proof. For $\lambda > 0$. Since the exponential function is increasing,

$$\mathbb{P}(X - \mu \geq t) = \mathbb{P}(\lambda(X - \mu) \geq \lambda t) \quad (1.9)$$

$$= \mathbb{P}(e^{\lambda(X - \mu)} \geq e^{\lambda t}). \quad (1.10)$$

Applying Markov's inequality to the nonnegative random variable $e^{\lambda(X - \mu)}$ gives

$$\mathbb{P}(X - \mu \geq t) \leq \frac{\mathbb{E}[e^{\lambda(X - \mu)}]}{e^{\lambda t}} = \phi(\lambda)e^{-\lambda t}. \quad (1.11)$$

Taking logarithms yields

$$\log \mathbb{P}(X - \mu \geq t) \leq \log \phi(\lambda) - \lambda t.$$

Because the inequality holds for every admissible λ , minimizing the right-hand side over $\lambda \in [0, b]$ gives the result. $\square$

Remark 1.2.4. The optimization over λ is essential. Different values of λ produce different valid bounds, and the infimum selects the sharpest one obtained from this argument. In applications, the main work is therefore to control the MGF and then solve a one-dimensional optimization problem.

1.3 Sub-Gaussian Random Variables

1.3.1 Gaussian Tail Bound

As a first example, consider a Gaussian random variable

$$X \sim N(\mu, \sigma^2).$$

Its moment generating function is

$$\mathbb{E}[e^{\lambda X}] = \exp\left(\mu\lambda + \frac{\sigma^2\lambda^2}{2}\right), \quad \lambda \in \mathbb{R}. \quad (1.12)$$

Applying the preceding lemma, the upper-tail bound is determined by

$$\inf_{\lambda>0} \left\{ \log \mathbb{E}[e^{\lambda(X-\mu)}] - \lambda t \right\} = \inf_{\lambda>0} \left\{ \frac{\sigma^2\lambda^2}{2} - \lambda t \right\}. \quad (1.13)$$

Here we used the fact that

$$X - \mu \sim N(0, \sigma^2).$$

The quadratic expression is minimized at

$$\lambda^* = \frac{t}{\sigma^2},$$

which gives

$$\inf_{\lambda>0} \left\{ \frac{\sigma^2\lambda^2}{2} - \lambda t \right\} = -\frac{t^2}{2\sigma^2}. \quad (1.14)$$

Therefore, for every $t > 0$,

$$\mathbb{P}(X - \mu \geq t) \leq \exp\left(-\frac{t^2}{2\sigma^2}\right). \quad (1.15)$$

(Normal distribution is light tail)

1.3.2 Definition and basic tail bounds

Definition 1.3.1 (Sub-Gaussian random variable). A random variable X with mean $\mu = \mathbb{E}[X]$ is *sub-Gaussian with parameter $\sigma > 0$* if

$$\mathbb{E}\left[e^{\lambda(X-\mu)}\right] \leq \exp\left(\frac{\sigma^2\lambda^2}{2}\right) \quad \text{for all } \lambda \in \mathbb{R}. \quad (1.16)$$

Remark 1.3.2. If X is σ -sub-Gaussian for some $\sigma > 0$, then X is also σ' -sub-Gaussian for some $\sigma' \geq \sigma$.

The definition says that the centered MGF of X grows no faster than the centered MGF of a Gaussian $N(0, \sigma^2)$. Therefore, the same Chernoff calculation gives the Gaussian-type upper-tail bound

$$\mathbb{P}(X - \mu \geq t) \leq \exp\left(-\frac{t^2}{2\sigma^2}\right), \quad t > 0. \quad (1.17)$$

Lemma 1.3.3 (Symmetry of sub-Gaussianity). *A random variable X is sub-Gaussian with parameter σ if and only if $-X$ is sub-Gaussian with the same parameter.*

Proof. Let $\mu = \mathbb{E}[X]$. Recall that X is sub-Gaussian with parameter σ if

$$\mathbb{E}\left[e^{\lambda(X-\mu)}\right] \leq \exp\left(\frac{\sigma^2\lambda^2}{2}\right), \quad \lambda \in \mathbb{R}. \quad (1.18)$$

Suppose first that X is sub-Gaussian. Since $\mathbb{E}[-X] = -\mu$,

$$\mathbb{E}\left[e^{\lambda((-X)-\mathbb{E}[-X])}\right] = \mathbb{E}\left[e^{-\lambda(X-\mu)}\right] \quad (1.19)$$

$$\leq \exp\left(\frac{\sigma^2(-\lambda)^2}{2}\right) \quad (1.20)$$

$$= \exp\left(\frac{\sigma^2\lambda^2}{2}\right). \quad (1.21)$$

Hence $-X$ is sub-Gaussian with parameter σ .

Conversely, suppose that $-X$ is sub-Gaussian. Then

$$\mathbb{E} \left[e^{\lambda(X-\mu)} \right] = \mathbb{E} \left[e^{-\lambda((-X)-\mathbb{E}[-X])} \right] \quad (1.22)$$

$$\leq \exp \left(\frac{\sigma^2(-\lambda)^2}{2} \right) \quad (1.23)$$

$$= \exp \left(\frac{\sigma^2\lambda^2}{2} \right). \quad (1.24)$$

Therefore X is sub-Gaussian with parameter σ . $\square$

Lemma 1.3.4 (Lower Tail). *If X is sub-Gaussian with parameter σ , then for every $t > 0$,*

$$\mathbb{P}(X - \mathbb{E}[X] \leq -t) \leq \exp \left(-\frac{t^2}{2\sigma^2} \right). \quad (1.25)$$

Proof. By the symmetry lemma, $-X$ is also sub-Gaussian with parameter σ . Moreover,

$$\mathbb{P}(X - \mathbb{E}[X] \leq -t) = \mathbb{P}(-X - \mathbb{E}[-X] \geq t) \quad (1.26)$$

$$\leq \exp \left(-\frac{t^2}{2\sigma^2} \right), \quad (1.27)$$

where the last step follows from the upper-tail bound applied to $-X$. $\square$

Now if we combine upper and lower tails with the union bound yields the two-sided concentration inequality we will get the following

Lemma 1.3.5. *If X is sub-Gaussian with parameter $\sigma > 0$,*

$$\mathbb{P}(|X - \mathbb{E}X| \geq t) \leq 2 \exp \left(-\frac{t^2}{2\sigma^2} \right), \quad t > 0. \quad (1.28)$$

Note: This much faster decay is one reason sub-Gaussian assumptions are central in non-asymptotic high-dimensional statistics [1].

Example 1.3.6 (Rademacher random variable). A Rademacher random variable ε takes values in $\{-1, +1\}$ equiprobably:

In particular, $\mathbb{E}[\varepsilon] = 0$. Its MGF is

$$\mathbb{E}[e^{\lambda\varepsilon}] = \frac{1}{2}e^\lambda + \frac{1}{2}e^{-\lambda} \quad (1.29)$$

$$= \frac{1}{2} \left(\sum_{k=0}^{\infty} \frac{\lambda^k}{k!} + \sum_{k=0}^{\infty} \frac{(-\lambda)^k}{k!} \right) \quad (1.30)$$

$$= \sum_{m=0}^{\infty} \frac{\lambda^{2m}}{(2m)!}, \quad (1.31)$$

where the odd-order terms cancel.

Since $(2m)! \geq 2^m m!$,

$$\mathbb{E}[e^{\lambda\varepsilon}] \leq \sum_{m=0}^{\infty} \frac{\lambda^{2m}}{2^m m!} \quad (1.32)$$

$$= e^{\lambda^2/2}. \quad (1.33)$$

Since $\mathbb{E}[\varepsilon] = 0$, it follows that

$$\mathbb{E}[e^{\lambda(\varepsilon - \mathbb{E}[\varepsilon])}] \leq e^{\lambda^2/2}.$$

Therefore, ε is sub-Gaussian with parameter $\sigma = 1$.

This example shows that sub-Gaussianity is not the same as Gaussianity. A Rademacher variable is discrete and bounded, yet its MGF is dominated by that of a standard Gaussian. The sub-Gaussian class is therefore broad enough to include many non-Gaussian random variables while retaining Gaussian-like tail behavior.

Note: "equiprobably" meaning

$$\mathbb{P}(\varepsilon = 1) = \mathbb{P}(\varepsilon = -1) = \frac{1}{2}. \quad (1.34)$$

Therefore,

$$\mathbb{E}[\varepsilon] = \frac{1}{2}(1) + \frac{1}{2}(-1) = 0. \quad (1.35)$$

1.3.3 Example of Sub-Gaussian Variables

Example 1.3.7 (Bounded random variables). Let X be zero-mean, and supported on some interval $[a, b]$. We will show that X is sub-Gaussian with parameter at most $\sigma = b - a$.

Letting X' be an independent copy of X , so that $X \stackrel{d}{=} X'$ and $X \perp\!\!\!\perp X'$. For any $\lambda \in \mathbb{R}$, we have

$$\mathbb{E}_X [e^{\lambda X}] = \mathbb{E}_X [e^{\lambda(X - \mathbb{E}_{X'}[X'])}] \stackrel{(i)}{\leq} \mathbb{E}_X [\mathbb{E}_{X'} e^{\lambda(X - X')}] = \mathbb{E}_{X, X'} [e^{\lambda(X - X')}]$$

The first equality is because $\mathbb{E}[X'] = 0$, and (i) follows from the the convexity of the exponential, and Jensen's inequality. Letting ε be an independent Rademacher variable, note that since (X, X') are i.i.d., $X - X' \stackrel{d}{=} X' - X$, so $(X - X')$ is symmetric about 0. Thus, the distribution of $(X - X')$ is the same as that of $\varepsilon(X - X')$, so that we have

$$\mathbb{E}_{X, X'} [e^{\lambda(X - X')}] = \mathbb{E}_{X, X'} [\mathbb{E}_\varepsilon [e^{\lambda\varepsilon(X - X')}]] \stackrel{(ii)}{\leq} \mathbb{E}_{X, X'} [e^{\frac{\lambda^2}{2}(X - X')^2}] \leq e^{\frac{\lambda^2}{2}(b-a)^2}$$

The first equality follows from Tonelli's theorem and (ii) follows from the sub-Gaussianity of Radamacher variable and last inequality is because $|X - X'| \leq b - a$. Putting everything together, we have shown that X is sub-Gaussian with parameter at most $\sigma = b - a$.

Remark 1.3.8. This result is useful, but it can be sharpened. It can be shown that X is sub-Gaussian with parameter at most $\sigma = \frac{b-a}{2}$ (Exercise 2.4). The technique used in this example is a simple example of a *symmetrization argument*, in which we first introduce an independent copy X' , and then symmetrize the problem with a Rademacher variable. Such symmetrization arguments are useful in a variety of contexts.

1.3.4 Hoeffding's Inequality

Another property of sub-Gaussianity is preserved by linear operations.

Lemma 1.3.9. *Let X_1 and X_2 be independent sub-Gaussian variables with parameters σ_1 and σ_2 , respectively, then $X_1 + X_2$ is sub-Gaussian with parameter $\sqrt{\sigma_1^2 + \sigma_2^2}$.*

Proof. By independence we have

$$\mathbb{E} [e^{\lambda(X_1 + X_2 - \mathbb{E}X_1 - \mathbb{E}X_2)}] = \mathbb{E} [e^{\lambda(X_1 - \mathbb{E}X_1)}] \mathbb{E} [e^{\lambda(X_2 - \mathbb{E}X_2)}] \leq e^{\frac{\lambda^2}{2}\sigma_1^2} e^{\frac{\lambda^2}{2}\sigma_2^2} = e^{\frac{\lambda^2}{2}(\sigma_1^2 + \sigma_2^2)}$$

□

As a consequence of this fact, we obtain an important result known as the *Hoeffding's inequality*:

Proposition 1.3.10 (Hoeffding's inequality). *Suppose that the variables X_i , $i = 1, \dots, n$, are independent, and X_i has mean μ_i and sub-Gaussian parameter σ_i . Then for all $t \geq 0$, we have*

$$\mathbb{P} \left[\sum_{i=1}^n (X_i - \mu_i) \geq t \right] \leq \exp \left\{ -\frac{t^2}{2 \sum_{i=1}^n \sigma_i^2} \right\}$$

and

$$\mathbb{P} \left[\left| \sum_{i=1}^n X_i - \sum_{i=1}^n \mathbb{E} X_i \right| \geq t \right] \leq 2 \exp \left\{ -\frac{t^2}{2 \sum_{i=1}^n \sigma_i^2} \right\}$$

Proof. By previous lemma, $\sum_{i=1}^n X_i$ is $\sqrt{\sigma_1^2 + \dots + \sigma_n^2}$ sub-Gaussian. So, by basic sub-Gaussian tail bound we have the results. $\square$

Corollary 1.3.11. *If X_i , $i = 1, \dots, n$, are i.i.d, with mean μ and sub-Gaussian parameter σ . Then for all $t \geq 0$, we have*

$$\mathbb{P} \left[\left| \frac{1}{n} \sum_{i=1}^n X_i - \mu \right| \geq t \right] \leq 2 \exp \left\{ -\frac{nt^2}{2\sigma^2} \right\}$$

and

$$\mathbb{P} \left[\sqrt{n} \left| \frac{1}{n} \sum_{i=1}^n X_i - \mu \right| \geq t \right] \leq 2 \exp \left\{ -\frac{t^2}{2\sigma^2} \right\}$$

Remark 1.3.12. The above can be viewed as a finite sample Central Limit Theorem. CLT states that when $n \rightarrow \infty$,

$$\sqrt{n} \left| \frac{1}{n} \sum_{i=1}^n X_i - \mu \right| \rightarrow \mathcal{N}(0, \sigma^2).$$

The second statement of the above corollary is a finite sample version of CLT, by stating that the tail of $\sqrt{n} \left| \frac{1}{n} \sum_{i=1}^n X_i - \mu \right|$ is upper bounded by the tail of $\mathcal{N}(0, \sigma^2)$.

Proof. By Hoeffding,

$$\mathbb{P} \left[\left| \frac{1}{n} \sum_{i=1}^n X_i - \mu \right| \geq t \right] = \mathbb{P} \left[\left| \sum_{i=1}^n X_i - n\mu \right| \geq nt \right] \leq 2 \exp \left\{ -\frac{(nt)^2}{2n\sigma^2} \right\} = 2 \exp \left\{ -\frac{nt^2}{2\sigma^2} \right\}$$

Second statement follows analogously. $\square$

One application of Hoeffding's bound is to derive finite sample guarantees for the error of the sample mean.

Example 1.3.13 (Finite-sample error guarantee). Let $\bar{X}_n := \frac{1}{n} \sum_{i=1}^n X_i$ and fix a confidence level $\delta \in (0, 1)$. By the preceding corollary,

$$\mathbb{P} \left[|\bar{X}_n - \mu| \geq t \right] \leq 2 \exp \left\{ -\frac{nt^2}{2\sigma^2} \right\}$$

Choose $t = \sqrt{\frac{2\sigma^2}{n} \log \left(\frac{2}{\delta} \right)}$, so that $\delta = 2 \exp \left\{ -\frac{nt^2}{2\sigma^2} \right\}$. In other words, with probability at least $1 - \delta$,

$$|\bar{X}_n - \mu| \leq \sqrt{\frac{2\sigma^2}{n} \log \left(\frac{2}{\delta} \right)}$$

Example 1.3.14 (Expected absolute error). We can also control the expected absolute error $\mathbb{E}|\bar{X}_n - \mu|$ directly from the tail bound. Recall that for any nonnegative random variable Y , Tonelli's theorem gives

$$\mathbb{E}[Y] = \int_0^\infty \mathbb{P}(Y \geq t) dt$$

Taking $Y = |\bar{X}_n - \mu|$, the preceding corollary and the trivial bound $\mathbb{P}(Y \geq t) \leq 1$ imply

$$\mathbb{P} [|\bar{X}_n - \mu| \geq t] \leq \min \left\{ 1, 2 \exp \left\{ -\frac{nt^2}{2\sigma^2} \right\} \right\}$$

Let t_0 be the point at which the two upper bounds agree. Thus $2e^{-\frac{nt_0^2}{2\sigma^2}} = 1$, which gives $t_0 = \sigma \sqrt{\frac{2 \log 2}{n}}$. Hence

$$\begin{aligned} \mathbb{E} |\bar{X}_n - \mu| &= \int_0^\infty \mathbb{P} [|\bar{X}_n - \mu| \geq t] dt \\ &\leq \int_0^{t_0} 1 dt + \underbrace{\int_{t_0}^\infty 2 \exp \left\{ -\frac{nt^2}{2\sigma^2} \right\} dt}_{z = \frac{\sqrt{n}}{\sigma} t, \quad dt = \frac{\sigma}{\sqrt{n}} dz} \\ &= t_0 + \frac{2\sigma}{\sqrt{n}} \int_{\frac{\sqrt{n}}{\sigma} t_0}^\infty e^{-z^2/2} dz \\ &= \frac{\sigma}{\sqrt{n}} \left(\sqrt{2 \log 2} + 2 \int_{\sqrt{2 \log 2}}^\infty e^{-z^2/2} dz \right) \\ &\leq 2 \frac{\sigma}{\sqrt{n}}. \end{aligned}$$

We conclude our discussion of sub-Gaussianity with a theorem that provides different characterizations of sub-Gaussian variables.

Theorem 1.3.15 (Equivalent characterizations of sub-Gaussian variables). *Given any zero-mean random variable X , the following properties are equivalent:*

1. *There is a constant $\sigma \geq 0$ such that*

$$\mathbb{E} [e^{\lambda X}] \leq e^{\frac{\lambda^2 \sigma^2}{2}} \quad \text{for all } \lambda \in \mathbb{R}.$$

2. *There is a constant $c \geq 0$ and Gaussian random variable $Z \sim \mathcal{N}(0, \tau^2)$ such that*

$$\mathbb{P} [|X| \geq s] \leq c \mathbb{P} [|Z| \geq s] \quad \text{for all } s \geq 0.$$

3. *There is a constant $\theta \geq 0$ such that*

$$\mathbb{E} [X^{2k}] \leq \frac{(2k)!}{2^k k!} \theta^{2k} \quad \text{for all } k = 1, 2, \dots$$

4. *There is a constant $\sigma \geq 0$ such that*

$$\mathbb{E} \left[e^{\frac{\lambda X^2}{2\sigma^2}} \right] \leq \frac{1}{\sqrt{1-\lambda}} \quad \text{for all } \lambda \in [0, 1).$$

Proof. See Appendix A (Section 2.4) in [1]. □

1.4 Symmetrization

Lemma 1.4.1. *Let $Y \perp\!\!\!\perp Z$ with $\mathbb{E}[Z] = 0$. Then for any convex function F ,*

$$\mathbb{E}[F(Y)] \leq \mathbb{E}[F(Y + Z)]. \quad (1.36)$$

Example 1.4.2. The function $F(x) = e^{\lambda x}$ is convex. Suppose $\mathbb{E}[X] = 0$. Applying Lemma 1.4.1 with $Y = X$ and $Z = -X'$, where $X' \perp\!\!\!\perp X$ and $X' \sim \text{Law}(X)$, we obtain

$$\mathbb{E}_X [e^{\lambda X}] \leq \mathbb{E}_{X, X'} [e^{\lambda(X - X')}] \quad (1.37)$$

Proof of Lemma 1.4.1. For every $y \in \mathbb{R}$, using $\mathbb{E}[Z] = 0$ and Jensen's inequality,

$$F(y) = F(y + \mathbb{E}_Z[Z]) = F(\mathbb{E}_Z[y + Z]) \leq \mathbb{E}_Z[F(y + Z)].$$

Substituting $y = Y$ gives $F(Y) \leq \mathbb{E}_Z[F(Y + Z)]$, and taking expectations over Y yields

$$\mathbb{E}_Y[F(Y)] \leq \mathbb{E}_Y \mathbb{E}_Z[F(Y + Z)] = \mathbb{E}_{Y, Z}[F(Y + Z)],$$

where the last equality uses $Y \perp\!\!\!\perp Z$. □

Definition 1.4.3. A random variable X is called *symmetric* if it has the same distribution as $-X$.

Remark 1.4.4. If X has pdf $f(x)$, then X is symmetric if and only if $f(x) = f(-x)$.

Lemma 1.4.5 (Constructing symmetric distributions). *Let X be a random variable and let ε be an independent Rademacher random variable.*

- (a) εX and $\varepsilon |X|$ are identically distributed and symmetric.
- (b) If X is symmetric, then both εX and $\varepsilon |X|$ have the same distribution as X .
- (c) If X' is an independent copy of X , then $X - X'$ is symmetric.

Proof. We only prove that εX is symmetric. For every event $A \subseteq \mathbb{R}$, conditioning on the value of ε gives

$$\mathbb{P}[\varepsilon X \in A] = \frac{1}{2} \mathbb{P}[X \in A] + \frac{1}{2} \mathbb{P}[-X \in A] = \frac{1}{2} \mathbb{P}[-X \in A] + \frac{1}{2} \mathbb{P}[X \in A] = \mathbb{P}[-\varepsilon X \in A]. \quad \square$$

Lemma 1.4.6 (Symmetrization). *Let $X_1, \dots, X_N$ be independent, mean-zero random vectors in some normed space, and let $\varepsilon_1, \dots, \varepsilon_N$ be independent Rademacher random variables. Then*

$$\frac{1}{2} \mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i X_i \right\| \leq \mathbb{E} \left\| \sum_{i=1}^N X_i \right\| \leq 2 \mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i X_i \right\|. \quad (1.38)$$

Proof. Let $(X'_1, \dots, X'_N)$ be independent copies of $(X_1, \dots, X_N)$. By Lemma 1.4.5(c) the vectors $X_i - X'_i$ are independent and symmetric (this step uses the assumption that X_i are independent), so that

$$(X_1 - X'_1, \dots, X_N - X'_N) \stackrel{d}{=} (\varepsilon_1(X_1 - X'_1), \dots, \varepsilon_N(X_N - X'_N)). \quad (1.39)$$

Upper bound. Set $p := \mathbb{E} \left\| \sum_{i=1}^N X_i \right\|$. Since $\mathbb{E} \left[\sum_{i=1}^N X'_i \right] = 0$, we may apply Lemma 1.4.1 with $F(x) = \|x\|$, $Y = \sum_{i=1}^N X_i$ and $Z = -\sum_{i=1}^N X'_i$ to get

$$p = \mathbb{E} \left\| \sum_{i=1}^N X_i - \mathbb{E} \left[\sum_{i=1}^N X'_i \right] \right\| \leq \mathbb{E} \left\| \sum_{i=1}^N (X_i - X'_i) \right\|.$$

Then by (1.39) and the triangle inequality,

$$\begin{aligned} \mathbb{E} \left\| \sum_{i=1}^N (X_i - X'_i) \right\| &= \mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i (X_i - X'_i) \right\| = \mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i X_i - \sum_{i=1}^N \varepsilon_i X'_i \right\| \\ &\leq \mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i X_i \right\| + \mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i X'_i \right\| = 2 \mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i X_i \right\|. \end{aligned}$$

Lower bound. Conditionally on $\varepsilon = (\varepsilon_1, \dots, \varepsilon_N)$, the vectors $\sum_{i=1}^N \varepsilon_i X_i$ and $-\sum_{i=1}^N \varepsilon_i X'_i$ are independent and the latter has mean zero, so Lemma 1.4.1 applies conditionally and gives, after averaging over ε ,

$$\mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i X_i \right\| \leq \mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i X_i - \sum_{i=1}^N \varepsilon_i X'_i \right\| = \mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i (X_i - X'_i) \right\|.$$

Applying (1.39) once more and then the triangle inequality,

$$\mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i (X_i - X'_i) \right\| = \mathbb{E} \left\| \sum_{i=1}^N (X_i - X'_i) \right\| \leq \mathbb{E} \left\| \sum_{i=1}^N X_i \right\| + \mathbb{E} \left\| \sum_{i=1}^N X'_i \right\| = 2 \mathbb{E} \left\| \sum_{i=1}^N X_i \right\|. \quad \square$$

Remark 1.4.7. The zero-mean assumption is needed for both bounds: in the upper bound to subtract $\mathbb{E}[\sum_i X'_i]$, and in the lower bound to guarantee that $\mathbb{E}[\sum_i \varepsilon_i X'_i \mid \varepsilon] = 0$.

1.5 Sub-exponential Random Variables

We now turn to the class of sub-exponential variables, which are defined by a slightly milder condition on the moment generating function.

Definition 1.5.1 ([1, Definition 2.7]). A random variable X with mean $\mu = \mathbb{E}[X]$ is *sub-exponential* if there are non-negative parameters (ν, α) such that

$$\mathbb{E} \left[e^{\lambda(X-\mu)} \right] \leq e^{\frac{\nu^2 \lambda^2}{2}} \quad \text{for all } |\lambda| < \frac{1}{\alpha}. \quad (1.40)$$

Remark 1.5.2. A σ -sub-Gaussian variable is $(\sigma, 0)$ -sub-exponential, where we interpret $1/0$ as $+\infty$.

Example 1.5.3 (Sub-exponential but not sub-Gaussian). Let $Z \sim N(0, 1)$ and consider $X = Z^2$, so that $\mathbb{E}[X] = 1$. For $\lambda < \frac{1}{2}$,

$$\begin{aligned} \mathbb{E} \left[e^{\lambda(X-1)} \right] &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{\lambda(z^2-1)} e^{-\frac{z^2}{2}} dz \\ &= \frac{e^{-\lambda}}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-\frac{(1-2\lambda)z^2}{2}} dz = \frac{e^{-\lambda}}{\sqrt{1-2\lambda}}, \end{aligned} \quad (1.41)$$

where the last equality uses $\int_{-\infty}^{+\infty} e^{-az^2/2} dz = \sqrt{2\pi/a}$ with $a = 1-2\lambda > 0$. For $\lambda \geq \frac{1}{2}$ the moment generating function is infinite, so X is not sub-Gaussian. On the other hand,

$$\frac{e^{-\lambda}}{\sqrt{1-2\lambda}} \leq e^{2\lambda^2} = e^{\frac{2^2 \lambda^2}{2}} \quad \text{where } |\lambda| < \frac{1}{4} \quad (\text{check this!}), \quad (1.42)$$

so that $\mathbb{E} \left[e^{\lambda(X-1)} \right] \leq e^{\frac{2^2 \lambda^2}{2}}$ for all $|\lambda| < \frac{1}{4}$, i.e. X is $(2, 4)$ -sub-exponential.

Exercises

1. Verify the inequality (1.42), that is, show that $\frac{e^{-\lambda}}{\sqrt{1-2\lambda}} \leq e^{2\lambda^2}$ for all $|\lambda| < \frac{1}{4}$.

Proposition 1.5.4 (Sub-exponential tail bound). *Suppose X is sub-exponential with parameters (ν, α) and mean μ . Then*

$$\mathbb{P}[X - \mu \geq t] \leq \begin{cases} e^{-\frac{t^2}{2\nu^2}}, & \text{if } 0 \leq t \leq \frac{\nu^2}{\alpha}, \\ e^{-\frac{t}{2\alpha}}, & \text{for } t > \frac{\nu^2}{\alpha}, \end{cases} = e^{-\min\left\{\frac{t^2}{2\nu^2}, \frac{t}{2\alpha}\right\}}. \quad (1.43)$$

Proof. Assume $\mu = 0$ without loss of generality. By the Chernoff bound from Lecture 1 and Definition 1.5.1,

$$\begin{aligned} \log \mathbb{P}[X \geq t] &\leq \inf_{0 \leq \lambda < \frac{1}{\alpha}} \left\{ \log \mathbb{E}[e^{\lambda X}] - \lambda t \right\} \leq \inf_{0 \leq \lambda < \frac{1}{\alpha}} \left\{ \log e^{\frac{\nu^2 \lambda^2}{2}} - \lambda t \right\} \\ &= \inf_{0 \leq \lambda < \frac{1}{\alpha}} \underbrace{\left\{ \frac{\nu^2 \lambda^2}{2} - \lambda t \right\}}_{=: g(\lambda)}. \end{aligned} \quad (1.44)$$

The function g is a convex parabola (see Figure 1.2) with

$$g'(\lambda) = \nu^2 \lambda - t = 0 \quad \Longleftrightarrow \quad \lambda = \frac{t}{\nu^2}.$$

- ① If $\frac{t}{\nu^2} < \frac{1}{\alpha}$ (i.e. $t < \frac{\nu^2}{\alpha}$), the unconstrained minimizer lies in the feasible region, so

$$\inf_{0 \leq \lambda < \frac{1}{\alpha}} g(\lambda) = g\left(\frac{t}{\nu^2}\right) = \frac{\nu^2}{2} \cdot \frac{t^2}{\nu^4} - \frac{t}{\nu^2} \cdot t = -\frac{t^2}{2\nu^2}.$$

- ② If $\frac{t}{\nu^2} \geq \frac{1}{\alpha}$ (i.e. $t \geq \frac{\nu^2}{\alpha}$), then g is decreasing on $[0, \frac{t}{\nu^2})$ and the infimum is approached at $\lambda = \frac{1}{\alpha}$, giving

$$\inf_{0 \leq \lambda < \frac{1}{\alpha}} g(\lambda) = g\left(\frac{1}{\alpha}\right) = \frac{\nu^2}{2} \cdot \frac{1}{\alpha^2} - \frac{1}{\alpha} t = \frac{\nu^2}{2\alpha^2} - \frac{t}{\alpha} \leq -\frac{t}{2\alpha},$$

where the last step uses $\nu^2 < \alpha t$. □

1.5.1 Bernstein's Condition

We say that X satisfies *Bernstein's condition with parameter b* if

$$|\mathbb{E}[(X - \mu)^k]| \leq \frac{1}{2} k! \operatorname{Var}(X) b^{k-2}, \quad \text{for any } k = 2, 3, \dots, \quad (1.45)$$

where $\mu = \mathbb{E}[X]$.

Example 1.5.5. If (1.45) is satisfied, then X is sub-exponential. Indeed, writing $\sigma^2 = \operatorname{Var}(X)$ and using the Taylor expansion of the exponential together with $\mathbb{E}[X - \mu] = 0$,

$$\begin{aligned} \mathbb{E}[e^{\lambda(X-\mu)}] &= \mathbb{E}\left[\sum_{k=0}^{\infty} \frac{(\lambda(X-\mu))^k}{k!}\right] = 1 + 0 + \mathbb{E}\left[\sum_{k=2}^{\infty} \frac{\lambda^k (X-\mu)^k}{k!}\right] \\ &= 1 + \sum_{k=2}^{\infty} \frac{\lambda^k \mathbb{E}[(X-\mu)^k]}{k!} = 1 + \frac{\lambda^2 \sigma^2}{2} + \sum_{k=3}^{\infty} \frac{\lambda^k \mathbb{E}[(X-\mu)^k]}{k!} \\ &\leq 1 + \frac{\lambda^2 \sigma^2}{2} + \sum_{k=3}^{\infty} \frac{|\lambda|^k \cdot \frac{1}{2} k! \sigma^2 b^{k-2}}{k!} = 1 + \frac{\lambda^2 \sigma^2}{2} + \frac{\lambda^2 \sigma^2}{2} \sum_{k=3}^{\infty} (|\lambda| b)^{k-2} = 1 + \frac{\lambda^2 \sigma^2}{2} \sum_{s=0}^{\infty} (|\lambda| b)^s. \end{aligned} \quad (1.46)$$

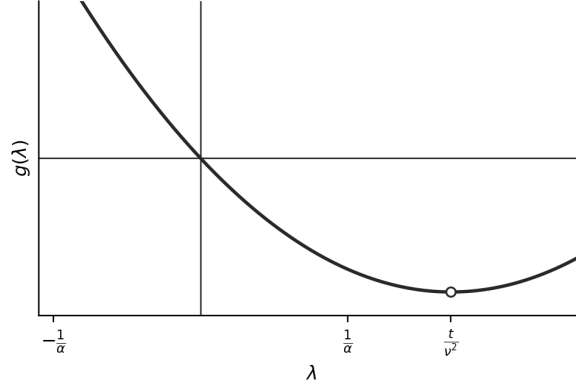


Figure 1.2: The function $g(\lambda) = \frac{\nu^2 \lambda^2}{2} - \lambda t$ appearing in (1.44), in the case $\frac{t}{\nu^2} > \frac{1}{\alpha}$. The open circle marks the unconstrained minimizer $\frac{t}{\nu^2}$, which lies outside the feasible region $|\lambda| < \frac{1}{\alpha}$.

For $|\lambda| < \frac{1}{b}$ the geometric series sums to $\frac{1}{1-|\lambda|b}$, so that

$$\mathbb{E} \left[e^{\lambda(X-\mu)} \right] \leq 1 + \frac{\lambda^2 \sigma^2 / 2}{1 - b|\lambda|} \leq e^{\frac{\lambda^2 \sigma^2 / 2}{1 - b|\lambda|}}, \quad (1.47)$$

using $1 + t \leq e^t$. Restricting further to $|\lambda| < \frac{1}{2b}$ gives $1 - b|\lambda| > \frac{1}{2}$, and therefore

$$\mathbb{E} \left[e^{\lambda(X-\mu)} \right] \leq e^{\lambda^2 \sigma^2} = e^{\frac{\lambda^2 (\sqrt{2}\sigma)^2}{2}}, \quad \text{for all } |\lambda| < \frac{1}{2b},$$

i.e. X is sub-exponential with parameters $(\nu, \alpha) = (\sqrt{2}\sigma, 2b)$.

Proposition 1.5.6 (Bernstein-type bound). *Suppose X satisfies Bernstein's condition (1.45) with parameter b , mean μ , and variance σ^2 . Then*

$$\mathbb{E} \left[e^{\lambda(X-\mu)} \right] \leq \exp \left(\frac{\lambda^2 \sigma^2 / 2}{1 - b|\lambda|} \right) \quad \text{for all } |\lambda| < \frac{1}{b}, \quad (1.48)$$

and, moreover, the concentration inequality

$$\mathbb{P}(|X - \mu| > t) \leq 2 \exp \left(-\frac{t^2}{2(\sigma^2 + bt)} \right) \quad \text{for all } t \geq 0. \quad (1.49)$$

Proof. Equation (1.48) directly follows from (1.46).

Remark 1.5.7. Restricting further to $|\lambda| < \frac{1}{2b}$ gives $b|\lambda| < \frac{1}{2}$, hence $1 - b|\lambda| > \frac{1}{2}$ and

$$\mathbb{E} \left[e^{\lambda X} \right] \leq \exp \left(\frac{\lambda^2 \sigma^2 / 2}{1/2} \right) = e^{\lambda^2 \sigma^2} = e^{\frac{\lambda^2 (\sqrt{2}\sigma)^2}{2}}, \quad |\lambda| < \frac{1}{2b},$$

which recovers the Topic 3 conclusion that X is $(\sqrt{2}\sigma, 2b)$ -sub-exponential. Bound (1.48) is sharper: it holds on the full range $|\lambda| < 1/b$ instead of only $|\lambda| < 1/(2b)$, and this extra room is exactly what lets us optimize the Chernoff bound below.

Proof of (1.49). By the Chernoff bound and (1.48), for any $t \geq 0$,

$$\log \mathbb{P}(X - \mu \geq t) \leq \inf_{0 \leq \lambda < 1/b} \left\{ \log \mathbb{E} \left[e^{\lambda(X - \mu)} \right] - \lambda t \right\} \leq \inf_{0 \leq \lambda < 1/b} \underbrace{\left\{ \frac{\lambda^2 \sigma^2}{2(1 - b\lambda)} - \lambda t \right\}}_{=: \theta(\lambda)}. \quad (1.50)$$

Rather than optimizing θ exactly, it suffices to evaluate it at the specific point

$$\lambda^* := \frac{t}{\sigma^2 + bt}.$$

Since $t, \sigma^2, b \geq 0$ we have $\lambda^* \geq 0$, and $\lambda^* < 1/b \iff bt < \sigma^2 + bt \iff \sigma^2 > 0$, which holds whenever X is non-degenerate (if $\sigma^2 = 0$ then $X \equiv \mu$ a.s. and (1.49) is trivial). At $\lambda = \lambda^*$,

$$1 - b\lambda^* = 1 - \frac{bt}{\sigma^2 + bt} = \frac{\sigma^2}{\sigma^2 + bt},$$

so that

$$\theta(\lambda^*) = \frac{\sigma^2(\lambda^*)^2(\sigma^2 + bt)}{2\sigma^2} - \lambda^* t = \frac{(\lambda^*)^2(\sigma^2 + bt)}{2} - \lambda^* t = \frac{t^2}{2(\sigma^2 + bt)} - \frac{t^2}{\sigma^2 + bt} = -\frac{t^2}{2(\sigma^2 + bt)}.$$

Combining with (1.50) (using that λ^* is feasible),

$$\mathbb{P}(X - \mu \geq t) \leq \exp \left(-\frac{t^2}{2(\sigma^2 + bt)} \right), \quad t \geq 0. \quad (1.51)$$

Finally, $-X$ also satisfies Bernstein's condition with the same (σ^2, b) , since $|\mathbb{E}[(-(X - \mu))^k]| = |\mathbb{E}[(X - \mu)^k]|$ and $\text{Var}(-X) = \sigma^2$. Applying (1.51) to $-X$ gives $\mathbb{P}(X - \mu \leq -t) \leq \exp(-t^2/(2(\sigma^2 + bt)))$ as well, and adding the two one-sided bounds yields (1.49). $\square$

Exercise 1.5.8. Compare the bound (1.49) with the bound one would obtain by plugging the (weaker) parameters $(\sqrt{2}\sigma, 2b)$ derived in the Remark above directly into the general sub-exponential tail bound of Topic 3. Show that (1.49) is never worse, and can be substantially tighter when $bt \ll \sigma^2$.

1.5.2 Sums of Independent Sub-exponential Random Variables

Recall that X with mean μ is *sub-exponential with parameters* (ν, α) if $\mathbb{E}[e^{\lambda(X - \mu)}] \leq e^{\nu^2 \lambda^2 / 2}$ for all $|\lambda| < 1/\alpha$, and that in this case

$$\mathbb{P}(|X - \mu| > t) \leq \begin{cases} 2e^{-t^2/(2\nu^2)}, & 0 \leq t \leq \nu^2/\alpha, \\ 2e^{-t/(2\alpha)}, & t > \nu^2/\alpha. \end{cases} \quad (1.52)$$

Lemma 1.5.9. Let $X_1, \dots, X_n$ be independent random variables, with X_i sub-exponential with parameters (ν_i, d_i) . Then $S_n := \sum_{i=1}^n X_i$ is sub-exponential with parameters

$$\left(\sqrt{\sum_{i=1}^n \nu_i^2}, \max_{1 \leq i \leq n} d_i \right).$$

and

$$\mathbb{P}(|S_n - \mathbb{E}[S_n]| > t) \leq \begin{cases} 2e^{-t^2/(2V^2)}, & 0 \leq t \leq V^2/d, \\ 2e^{-t/(2d)}, & t > V^2/d, \end{cases} \quad V^2 = \sum_{i=1}^n \nu_i^2, \quad d = \max_{1 \leq i \leq n} d_i. \quad (1.53)$$

Proof. Write $d := \max_{1 \leq i \leq n} d_i$ and $V^2 := \sum_{i=1}^n \nu_i^2$. For any $|\lambda| < 1/d = \min_{1 \leq i \leq n} (1/d_i)$, we have $|\lambda| < 1/d_i$ for every i , so by independence,

$$\mathbb{E} \left[e^{\lambda(S_n - \mathbb{E}[S_n])} \right] = \mathbb{E} \left[e^{\lambda \sum_{i=1}^n (X_i - \mathbb{E}[X_i])} \right] = \prod_{i=1}^n \mathbb{E} \left[e^{\lambda(X_i - \mathbb{E}[X_i])} \right] \leq \prod_{i=1}^n e^{\nu_i^2 \lambda^2 / 2} = e^{\lambda^2 V^2 / 2}.$$

Hence S_n is (V, d) -sub-exponential.

By (1.52) applied with $(\nu, \alpha) = (V, d)$, we obtain the tail bound

$$\mathbb{P}(|S_n - \mathbb{E}[S_n]| > t) \leq \begin{cases} 2e^{-t^2/(2V^2)}, & 0 \leq t \leq V^2/d, \\ 2e^{-t/(2d)}, & t > V^2/d, \end{cases} \quad V^2 = \sum_{i=1}^n \nu_i^2, \quad d = \max_{1 \leq i \leq n} d_i.$$

□

Example 1.5.10 (Concentration of χ^2 variables). Let $Z_1, \dots, Z_n$ i.i.d. $N(0, 1)$ and let $Y := \sum_{i=1}^n Z_i^2 \sim \chi_n^2$, so that $\mathbb{E}[Y] = n$. Recall from Topic 3 that each Z_i^2 has mean 1 and is $(2, 4)$ -sub-exponential, i.e. $\mathbb{E}[e^{\lambda(Z_i^2 - 1)}] \leq e^{2\lambda^2}$ for all $|\lambda| < 1/4$. Applying Lemma 1.5.9 with $\nu_i = 2$, $d_i = 4$ for every i , Y is $(\sqrt{4n}, 4) = (2\sqrt{n}, 4)$ -sub-exponential; here $V^2 = 4n$ and $d = 4$, so the threshold in (1.53) is $V^2/d = n$. Thus

$$\mathbb{P}(|Y - n| \geq s) \leq \begin{cases} 2e^{-s^2/(8n)}, & 0 \leq s \leq n, \\ 2e^{-s/8}, & s > n. \end{cases}$$

Substituting $s = nt$ (so $s \leq n \iff t \leq 1$) gives

$$\mathbb{P}\left(\left|\frac{Y}{n} - 1\right| \geq t\right) \leq \begin{cases} 2e^{-nt^2/8}, & 0 \leq t \leq 1, \\ 2e^{-nt/8}, & t > 1. \end{cases} \quad (1.54)$$

1.5.3 Application: The Johnson–Lindenstrauss Lemma

Fix distinct points $u^1, \dots, u^N \in \mathbb{R}^d$, where d is large. We would like a map $F : \mathbb{R}^d \rightarrow \mathbb{R}^m$ with $m \ll d$, so that we only need to store the compressed points $\{F(u^1), \dots, F(u^N)\}$, while approximately preserving all pairwise (squared) distances: given a distortion parameter $\delta \in (0, 1)$, we want

$$1 - \delta \leq \frac{\|F(u^i) - F(u^j)\|_2^2}{\|u^i - u^j\|_2^2} \leq 1 + \delta, \quad \text{for all } i \neq j. \quad (1.55)$$

Construction. Let $X \in \mathbb{R}^{m \times d}$ have i.i.d. $N(0, 1)$ entries, and write its rows as $X_1, \dots, X_m \in \mathbb{R}^d$, so that $X_1, \dots, X_m$ i.i.d. $N(0, I_d)$. Define the (random) linear map

$$F(u) := \frac{Xu}{\sqrt{m}}, \quad u \in \mathbb{R}^d.$$

Step 1: distribution of $\|Xu\|_2^2$. Fix $u \in \mathbb{R}^d$, $u \neq 0$. Writing $Xu = (\langle X_1, u \rangle, \dots, \langle X_m, u \rangle)^\top$, for each i and any unit vector $v = u/\|u\|_2$,

$$\langle X_i, v \rangle = \sum_{k=1}^d X_{i,k} v_k \sim N\left(0, \sum_{k=1}^d v_k^2\right) = N(0, 1) \quad (\text{check: } \|v\|_2 = 1), \quad (1.56)$$

since $X_i \sim N(0, I_d)$ and any linear combination of independent standard normals with coefficient vector of unit ℓ_2 -norm is again standard normal. Moreover $\langle X_1, v \rangle, \dots, \langle X_m, v \rangle$ are independent because $X_1, \dots, X_m$ are independent. Hence

$$Y := \frac{\|Xu\|_2^2}{\|u\|_2^2} = \sum_{i=1}^m \frac{\langle X_i, u \rangle^2}{\|u\|_2^2} = \sum_{i=1}^m \langle X_i, v \rangle^2 \sim \chi_m^2. \quad (1.57)$$

Step 2: concentration for a single vector. Since $\|F(u)\|_2^2 / \|u\|_2^2 = \|Xu\|_2^2 / (m \|u\|_2^2) = Y/m$, applying (1.54) (with n replaced by m) gives, for any $\delta \in (0, 1)$,

$$\mathbb{P} \left(\frac{\|F(u)\|_2^2}{\|u\|_2^2} \notin (1 - \delta, 1 + \delta) \right) = \mathbb{P} \left(\left| \frac{Y}{m} - 1 \right| \geq \delta \right) \leq 2e^{-m\delta^2/8}. \quad (1.58)$$

Step 3: union bound over all pairs. Apply (1.58) with $u = u^i - u^j$ for each of the $\binom{N}{2}$ unordered pairs $i \neq j$ (assuming $u^i \neq u^j$):

$$\mathbb{P} \left(\frac{\|F(u^i) - F(u^j)\|_2^2}{\|u^i - u^j\|_2^2} \notin (1 - \delta, 1 + \delta) \right) \leq 2e^{-m\delta^2/8}.$$

Let

$$I := \bigcup_{i < j} \left\{ \frac{\|F(u^i) - F(u^j)\|_2^2}{\|u^i - u^j\|_2^2} \notin (1 - \delta, 1 + \delta) \right\}$$

be the event that (1.55) fails for at least one pair. By the union bound,

$$\mathbb{P}(I) \leq \sum_{i < j} \mathbb{P} \left(\frac{\|F(u^i) - F(u^j)\|_2^2}{\|u^i - u^j\|_2^2} \notin (1 - \delta, 1 + \delta) \right) \leq \binom{N}{2} \cdot 2e^{-m\delta^2/8} = N(N-1)e^{-m\delta^2/8}. \quad (1.59)$$

Step 4: choosing m . Given a target failure probability $\varepsilon \in (0, 1)$, we want $\mathbb{P}(I) < \varepsilon$. By (1.59), it suffices that

$$N(N-1)e^{-m\delta^2/8} < \varepsilon \iff m > \frac{8}{\delta^2} \log \frac{N(N-1)}{\varepsilon}.$$

Since $N(N-1) \leq N^2$, a cleaner sufficient condition is

$$m > \frac{16}{\delta^2} \log \frac{N}{\varepsilon}. \quad (1.60)$$

Theorem 1.5.11 (Johnson–Lindenstrauss Lemma). *Let distinct $u^1, \dots, u^N \in \mathbb{R}^d$ and let $\delta, \varepsilon \in (0, 1)$. If*

$$m > \frac{16}{\delta^2} \log \frac{N}{\varepsilon},$$

then, with $X \in \mathbb{R}^{m \times d}$ having i.i.d. $N(0, 1)$ entries and $F(u) := Xu/\sqrt{m}$, with probability at least $1 - \varepsilon$ we have

$$1 - \delta \leq \frac{\|F(u^i) - F(u^j)\|_2^2}{\|u^i - u^j\|_2^2} \leq 1 + \delta \quad \text{simultaneously for all } i \neq j.$$

Proof. Immediate from (1.59) and (1.60): the choice of m forces $\mathbb{P}(I) < \varepsilon$, i.e. (1.55) holds for every pair on the complementary event, which has probability at least $1 - \varepsilon$. $\square$

Theorem 1.5.12 (Equivalent characterizations of sub-exponential random variables). *For a mean-zero random variable X , the following conditions are equivalent.*

(i) *There exist non-negative parameters (ν, α) such that*

$$\mathbb{E} [e^{\lambda X}] \leq e^{\nu^2 \lambda^2 / 2} \quad \text{for all } |\lambda| \leq \frac{1}{\alpha}.$$

(ii) *There exist constants $c_0, c_0 > 0$ such that*

$$\mathbb{E} [e^{\lambda X}] \leq c_0 \quad \text{for all } |\lambda| \leq c_0.$$

(iii) There exist constants $c_1, c_2 > 0$ such that

$$\mathbb{P}(|X| \geq t) \leq c_1 e^{-c_2 t} \quad \text{for all } t \geq 0.$$

(iv) The quantity

$$\gamma := \sup_{k \geq 2} \left(\frac{\mathbb{E}[|X|^k]}{k!} \right)^{1/k}$$

is finite.

Remark 1.5.13. A sub-exponential tail bound decays more slowly than a sub-Gaussian tail bound for large t , so the sub-exponential class permits heavier tails.

1.6 Martingale-Based Methods

Definition 1.6.1 (Martingale difference sequence). An adapted sequence $\{(D_k, \mathcal{F}_k)\}_{k=0}^\infty$ is called a *martingale difference sequence* if, for every $k \geq 1$,

$$\mathbb{E}[|D_k|] < \infty \quad \text{and} \quad \mathbb{E}[D_k \mid \mathcal{F}_{k-1}] = 0.$$

Equivalently, after shifting the index, $\mathbb{E}[D_{k+1} \mid \mathcal{F}_k] = 0$.

Remark 1.6.2. Here, D_k denotes the difference and $\mathcal{F}_k$ denotes the associated sigma-algebra.

Example 1.6.3. Let $\{(Y_k, \mathcal{F}_k)\}_{k \geq 0}$ be a martingale and define

$$D_k := Y_k - Y_{k-1}, \quad k \geq 1.$$

Then D_k is $\mathcal{F}_k$ -measurable and

$$\mathbb{E}[D_{k+1} \mid \mathcal{F}_k] = \mathbb{E}[Y_{k+1} \mid \mathcal{F}_k] - Y_k = 0 \quad \text{a.s.}$$

Moreover,

$$Y_k - Y_0 = \sum_{i=1}^k D_i.$$

Theorem 1.6.4. Let $\{(D_k, \mathcal{F}_k)\}$ be a martingale difference sequence. Suppose that, for every $k = 1, \dots, n$,

$$\mathbb{E}[e^{\lambda D_k} \mid \mathcal{F}_{k-1}] \leq e^{\lambda^2 \nu_k^2 / 2} \quad \text{for all } |\lambda| < \frac{1}{\alpha_k}.$$

Define

$$\alpha_* := \max_{1 \leq k \leq n} \alpha_k.$$

Then the following statements hold.

(a) $\sum_{k=1}^n D_k$ is sub-exponential with parameters $\left(\sqrt{\sum_{k=1}^n \nu_k^2}, \alpha_*\right)$.

(b) $\mathbb{P}(|\sum_{k=1}^n D_k| \geq t) \leq \begin{cases} 2 \exp\left(-\frac{t^2}{2 \sum_{k=1}^n \nu_k^2}\right), & 0 \leq t \leq \frac{\sum_{k=1}^n \nu_k^2}{\alpha_*}, \\ 2 \exp\left(-\frac{t}{2\alpha_*}\right), & t > \frac{\sum_{k=1}^n \nu_k^2}{\alpha_*}. \end{cases}$

Remark 1.6.5. The moment-generating-function assumption in Theorem 1.6.4 is a conditional sub-exponential condition. Part (a) is the martingale-difference analogue of the standard result that a sum of independent sub-exponential random variables is sub-exponential.

Proof. First, the tower property gives

$$\mathbb{E}[D_k] = \mathbb{E}[\mathbb{E}[D_k \mid \mathcal{F}_{k-1}]] = 0,$$

and hence $\mathbb{E}[\sum_{k=1}^n D_k] = 0$.

For $|\lambda| < 1/\alpha_*$, condition successively on the filtration:

$$\begin{aligned} \mathbb{E}\left[e^{\lambda \sum_{k=1}^n D_k}\right] &= \mathbb{E}\left[\mathbb{E}\left[e^{\lambda \sum_{k=1}^n D_k} \mid \mathcal{F}_{n-1}\right]\right] \\ &= \mathbb{E}\left[e^{\lambda \sum_{k=1}^{n-1} D_k} \mathbb{E}\left[e^{\lambda D_n} \mid \mathcal{F}_{n-1}\right]\right] \\ &\leq \mathbb{E}\left[e^{\lambda \sum_{k=1}^{n-1} D_k} e^{\lambda^2 \nu_n^2/2}\right], \quad |\lambda| < \frac{1}{\alpha_n} \\ &= e^{\lambda^2 \nu_n^2/2} \mathbb{E}\left[e^{\lambda \sum_{k=1}^{n-1} D_k}\right], \quad |\lambda| < \frac{1}{\alpha_n} \\ &\leq e^{\lambda^2 \nu_n^2/2} e^{\lambda^2 \nu_{n-1}^2/2} \mathbb{E}\left[e^{\lambda \sum_{k=1}^{n-2} D_k}\right], \quad |\lambda| < \frac{1}{\alpha_n}, |\lambda| < \frac{1}{\alpha_{n-1}} \\ &\leq \dots \leq e^{\lambda^2 \nu_n^2/2} \dots e^{\lambda^2 \nu_1^2/2}, \quad |\lambda| < \frac{1}{\alpha_n}, \dots, |\lambda| < \frac{1}{\alpha_1} \\ &= \exp\left(\frac{\lambda^2}{2} \sum_{k=1}^n \nu_k^2\right), \quad |\lambda| < \frac{1}{\max_{1 \leq k \leq n} \alpha_k}. \end{aligned}$$

This proves part (a). Part (b) follows from the standard two-regime sub-exponential tail bound. $\square$

Remark 1.6.6. If the D_k were independent, the MGF would factor as $\prod_{k=1}^n \mathbb{E}[e^{\lambda D_k}]$. For a martingale difference sequence, conditional expectation replaces this factorization. Given $\mathcal{F}_{n-1}$, the partial sum $\sum_{k=1}^{n-1} D_k$ is $\mathcal{F}_{n-1}$ -measurable and therefore acts as a constant inside the conditional expectation.

1.6.1 Azuma–Hoeffding Inequality

Corollary 1.6.7 (Azuma–Hoeffding random interval). *Let $\{(D_k, \mathcal{F}_k)\}_{k=1}^\infty$ be a martingale difference sequence and suppose $\mathbb{E}[D_k \mid \mathcal{F}_{k-1}] \in [a_k, b_k]$, where $a_k, b_k \in \mathcal{F}_{k-1}$ for all $k = 1, \dots, n$. Assume that $b_k - a_k \leq L_k$ is a constant. Then, for all $t \geq 0$,*

$$\mathbb{P}\left(\left|\sum_{k=1}^n D_k\right| \geq t\right) \leq 2 \exp\left(-\frac{2t^2}{\sum_{k=1}^n L_k^2}\right). \quad (1.61)$$

Proof. Conditioned on $\mathcal{F}_{k-1}$, the random variable D_k still lies in $[a_k, b_k]$ almost surely. By the conditional form of Hoeffding's lemma, D_k is conditionally $(b_k - a_k)/2$ -sub-Gaussian, and therefore $L_k/2$ -sub-Gaussian and is therefore conditionally sub-exponential with parameters

$$\left(\frac{L_k}{2}, 0\right).$$

By Theorem 1.6.4(a),

$$\sum_{k=1}^n D_k \text{ is } \left(\sqrt{\sum_{k=1}^n \frac{(L_k)^2}{4}}, 0\right) \text{ sub-exponential } \longrightarrow \left(\sqrt{\sum_{k=1}^n \frac{(L_k)^2}{4}}\right) \text{ sub-Gaussian.}$$

The stated tail bound follows. $\square$

Corollary 1.6.8 (Azuma–Hoeffding fix interval). *Let $\{(D_k, \mathcal{F}_k)\}$ be a martingale difference sequence. Suppose there are constants $\{(a_k, b_k)\}_{k=1}^\infty$ such that*

$$D_k \in [a_k, b_k] \quad \text{a.s. for every } k.$$

Then, for every $t \geq 0$,

$$\mathbb{P} \left(\left| \sum_{k=1}^n D_k \right| \geq t \right) \leq 2 \exp \left(- \frac{2t^2}{\sum_{k=1}^n (b_k - a_k)^2} \right).$$

Proof. It directly follows from the above corollary. $\square$

Remark 1.6.9. Conditional expectation preserves inequalities: if $X \leq b$, then $\mathbb{E}[X \mid \mathcal{F}] \leq \mathbb{E}[b \mid \mathcal{F}] = b$. Also, every sub-Gaussian random variable is sub-exponential.

1.6.2 Bounded Differences

Definition 1.6.10 (Bounded difference property). We say that a function f satisfies the bounded difference property with parameters $(L_1, \dots, L_n)$ if, for every $k = 1, \dots, n$,

$$|f(x_1, \dots, x_n) - f(x_1, \dots, x_{k-1}, x'_k, x_{k+1}, \dots, x_n)| \leq L_k$$

for all $x_1, \dots, x_n, x'_k$.

Example 1.6.11. For $x, y \in \mathbb{R}^n$, define the Hamming distance by

$$d_H(x, y) := \sum_{i=1}^n \mathbf{1}\{x_i \neq y_i\}.$$

If f is L -Lipschitz with respect to d_H , meaning that

$$|f(x) - f(y)| \leq L d_H(x, y),$$

then changing only the k th coordinate gives

$$|f(x_1, \dots, x_n) - f(x_1, \dots, x_{k-1}, x'_k, x_{k+1}, \dots, x_n)| \leq L \cdot 1 = L.$$

Thus, f has the bounded difference property with $L_k = L$ for every k .

Corollary 1.6.12 (Bounded difference inequality). *Suppose f satisfies the bounded difference property with parameters $(L_1, \dots, L_n)$ and that $X = (X_1, \dots, X_n)$ has independent components. Then, for every $t \geq 0$,*

$$\mathbb{P}(|f(X) - \mathbb{E}[f(X)]| \geq t) \leq 2 \exp \left(- \frac{2t^2}{\sum_{k=1}^n L_k^2} \right).$$

Proof. Let

$$\mathcal{F}_k := \sigma(X_1, \dots, X_k), \quad Y_k := \mathbb{E}[f(X) \mid \mathcal{F}_k].$$

Since $\mathcal{F}_n$ is generated by all the components of X ,

$$Y_n = \mathbb{E}[f(X) \mid \mathcal{F}_n] = f(X),$$

while

$$Y_0 = \mathbb{E}[f(X) \mid \mathcal{F}_0] = \mathbb{E}[f(X)].$$

Therefore,

$$f(X) - \mathbb{E}[f(X)] = Y_n - Y_0 = \sum_{k=1}^n D_k, \quad \text{where } D_k := Y_k - Y_{k-1}.$$

Note

$$\begin{aligned} D_k &= \mathbb{E}[f(X) \mid X_1, \dots, X_k] - \mathbb{E}[f(X) \mid X_1, \dots, X_{k-1}] \\ &= \mathbb{E}_{X_{k+1}, \dots, X_n}[f(X)] - \mathbb{E}_{X_k, X_{k+1}, \dots, X_n}[f(X)] \\ &:= \mathbb{E}_{k+1}[f(X)] - \mathbb{E}_k[f(X)]. \end{aligned}$$

Then $D_k \in \mathcal{F}_k$, and

$$\begin{aligned} \mathbb{E}[D_k \mid \mathcal{F}_{k-1}] &= \mathbb{E}[\mathbb{E}[f(X) \mid \mathcal{F}_k] - \mathbb{E}[f(X) \mid \mathcal{F}_{k-1}] \mid \mathcal{F}_{k-1}] \\ &= \mathbb{E}[f(X) \mid \mathcal{F}_{k-1}] - \mathbb{E}[f(X) \mid \mathcal{F}_{k-1}] \\ &= 0 \quad \text{a.s.} \end{aligned}$$

Thus, $(D_k)_{k=1}^n$ is a martingale difference sequence.

Next,

$$\begin{aligned} D_k &= \mathbb{E}_{k+1}[f(X_1, \dots, X_{k-1}, X_k, X_{k+1}, \dots, X_n)] - \mathbb{E}_k[f(X_1, \dots, X_{k-1}, X_k, X_{k+1}, \dots, X_n)] \\ &\leq \sup_x \mathbb{E}_{k+1}[f(X_1, \dots, X_{k-1}, x, X_{k+1}, \dots, X_n)] - \mathbb{E}_k[f(X_1, \dots, X_{k-1}, X_k, X_{k+1}, \dots, X_n)] \\ &:= B_k, \end{aligned}$$

where $B_k \in \mathcal{F}_{k-1}$. Similarly,

$$\begin{aligned} D_k &\geq \inf_x \mathbb{E}_{k+1}[f(X_1, \dots, X_{k-1}, x, X_{k+1}, \dots, X_n)] - \mathbb{E}_k[f(X_1, \dots, X_{k-1}, X_k, X_{k+1}, \dots, X_n)] \\ &:= A_k, \end{aligned}$$

where $A_k \in \mathcal{F}_{k-1}$. Then, we have

$$\begin{aligned} B_k - A_k &= \sup_x \mathbb{E}_{k+1}[f(X_1, \dots, X_{k-1}, x, X_{k+1}, \dots, X_n)] - \inf_{x'} \mathbb{E}_{k+1}[f(X_1, \dots, X_{k-1}, x', X_{k+1}, \dots, X_n)] \\ &\leq \sup_{x_1, \dots, x_{k-1}, x_k, x', x_{k+1}, \dots, x_n} |f(x_1, \dots, x_k, \dots, x_n) - f(x_1, \dots, x_{k-1}, x'_k, x_{k+1}, \dots, x_n)| \\ &\leq L_k, \end{aligned}$$

using (??). Therefore, by Azuma–Hoeffding with random interval,

$$\mathbb{P}\left(\left|\sum_{k=1}^n D_k\right| \geq t\right) = \mathbb{P}(|f(X) - \mathbb{E}f(X)| \geq t) \leq 2 \exp\left(-\frac{2t^2}{\sum_{k=1}^n L_k^2}\right).$$

□

Example 1.6.13 (U-statistics). Let $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a symmetric function of its arguments. Given i.i.d. sequence X_k , $k \geq 1$ of random variables, the quantity

$$U := \frac{1}{\binom{n}{2}} \sum_{1 \leq j < k \leq n} g(X_j, X_k) \quad (1.62)$$

is known as a pairwise U-statistic.

For example, suppose we want to estimate the mean absolute pairwise deviation $\mathbb{E}[|X_1 - X_2|]$. Then U with $g(x, y) = |x - y|$ is one estimate for it.

Since the X_i are identically distributed,

$$\mathbb{E}U = \frac{1}{\binom{n}{2}} \sum_{1 \leq j < k \leq n} \mathbb{E}g(X_j, X_k) = \frac{1}{\binom{n}{2}} \sum_{1 \leq j < k \leq n} \mathbb{E}g(X_1, X_2) = \mathbb{E}g(X_1, X_2).$$

Suppose $\|g\|_\infty \leq b$. Consider $U = f(X_1, \dots, X_n)$. Then

$$\begin{aligned}
& |f(X_1, \dots, X_n) - f(X_1, \dots, X_{k-1}, X'_k, X_{k+1}, \dots, X_n)| \\
&= \left| \frac{1}{\binom{n}{2}} \sum_{1 \leq j < s \leq n} g(X_j, X_s) - \frac{1}{\binom{n}{2}} \sum_{1 \leq j < s \leq n} g(X'_j, X'_s) \right| \quad [\text{where } X'_j = X_j, j = 1, \dots, k-1, k+1, \dots, n], \\
&= \frac{1}{\binom{n}{2}} \left| \sum_{1 \leq j < s \leq n} g(X_j, X_s) - \sum_{1 \leq j < s \leq n} g(X'_j, X'_s) \right| \\
&\leq \frac{1}{\binom{n}{2}} \sum_{1 \leq j < s \leq n} |g(X_j, X_s) - g(X'_j, X'_s)| \\
&\leq \frac{1}{\binom{n}{2}} |2b(n-k+k-1)| \\
&= \frac{4b}{n}.
\end{aligned}$$

So the bounded differences property holds with parameter $L_k = \frac{4b}{n}$ in each coordinate. Therefore, the bounded difference inequality gives

$$\mathbb{P}(|U - \mathbb{E}U| \geq t) \leq 2 \exp\left(-\frac{2t^2}{\sum_{k=1}^n (4b/n)^2}\right) = 2 \exp\left(-\frac{nt^2}{8b^2}\right). \quad (1.63)$$

Example 1.6.14 (Rademacher Complexity). Let $\{\varepsilon_k\}_{k=1}^n$ be an i.i.d. sequence of Rademacher variables. Given a collection of vectors $\mathcal{A} \subset \mathbb{R}^n$, define the random variable

$$Z(\mathcal{A}) = Z := \sup_{a \in \mathcal{A}} \langle a, \varepsilon \rangle = \sup_{a \in \mathcal{A}} \sum_{i=1}^n a_i \varepsilon_i. \quad (1.64)$$

Then $R(\mathcal{A}) := \mathbb{E}Z(\mathcal{A})$ is the Rademacher complexity of $\mathcal{A}$.

Now let us establish $Z(\mathcal{A}) = f(\varepsilon_1, \dots, \varepsilon_n)$ is sub-Gaussian. Given two Rademacher vectors $\varepsilon, \varepsilon' \in \{-1, +1\}^n$, and $\varepsilon'_i = \varepsilon_i$ for $i \neq k$, we have

$$\begin{aligned}
|f(\varepsilon_1, \dots, \varepsilon_n) - f(\varepsilon'_1, \dots, \varepsilon'_n)| &= \left| \sup_{a \in \mathcal{A}} \langle a, \varepsilon \rangle - \sup_{a \in \mathcal{A}} \langle a, \varepsilon' \rangle \right| \\
&\leq \sup_{a \in \mathcal{A}} |\langle a, \varepsilon \rangle - \langle a, \varepsilon' \rangle| \quad (\text{by Lemma 1.6.15}) \\
&= \sup_{a \in \mathcal{A}} |a_k \varepsilon_k - a_k \varepsilon'_k| \\
&= \sup_{a \in \mathcal{A}} |a_k| |\varepsilon_k - \varepsilon'_k| \\
&\leq 2 \sup_{a \in \mathcal{A}} |a_k|.
\end{aligned}$$

Consequently, the bounded difference inequality yields

$$\mathbb{P}(|Z(\mathcal{A}) - R(\mathcal{A})| \geq t) \leq 2 \exp\left(-\frac{2t^2}{\sum_{k=1}^n 4 \sup_{a \in \mathcal{A}} a_k^2}\right) = 2 \exp\left(-\frac{t^2}{2 \sum_{k=1}^n \sup_{a \in \mathcal{A}} a_k^2}\right).$$

Thus, $Z(\mathcal{A})$ is sub-Gaussian with parameter $\sqrt{2 \sum_{k=1}^n \sup_{a \in \mathcal{A}} a_k^2}$.

Lemma 1.6.15. *The absolute difference of the suprema of two functions is bounded by the supremum of their pointwise absolute difference*

$$\left| \sup_{x \in B} f(x) - \sup_{x \in B} g(x) \right| \leq \sup_{x \in B} |f(x) - g(x)|.$$

1.7 Lipschitz Functions of Gaussian Variables

We first show that Lipschitz functions of a standard Gaussian vector are sub-Gaussian.

Definition 1.7.1. A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is L -Lipschitz with respect to the Euclidean norm $\|\cdot\|_2$ if

$$|f(x) - f(y)| \leq L \|x - y\|_2 \quad \text{for all } x, y \in \mathbb{R}^n. \quad (1.65)$$

Theorem 1.7.2 ([1, Theorem 2.26]). Let $X = (X_1, \dots, X_n) \sim N(0, I_n)$, and let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be L -Lipschitz with respect to $\|\cdot\|_2$. Then $f(X) - \mathbb{E}[f(X)]$ is sub-Gaussian with parameter at most L , and hence

$$\mathbb{P}(|f(X) - \mathbb{E}[f(X)]| \geq t) \leq 2e^{-\frac{t^2}{2L^2}} \quad \text{for all } t \geq 0. \quad (1.66)$$

Remark 1.7.3. The sub-Gaussian parameter in Theorem 1.7.2 depends on f only through L and does not depend on the dimension n . Thus an L -Lipschitz function of an n -dimensional standard Gaussian vector obeys the same tail bound as a scalar $N(0, L^2)$ variable, however large n is. The tail bound (1.66) follows from the sub-Gaussian property by applying the Chernoff bound from Lecture 1 to $f(X)$ and to $-f(X)$.

We do not prove Theorem 1.7.2 here; a proof of the weaker statement with sub-Gaussian parameter $\pi L/2$ can be found in [1, Section 2.3]. We now consider three applications.

Example 1.7.4 (Order statistics, [1, Example 2.29]). Given a random vector $(X_1, \dots, X_n)$, its order statistics are obtained by reordering its entries in non-decreasing order,

$$X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}.$$

In particular, $X_{(n)} = \max_{1 \leq i \leq n} X_i$ and $X_{(1)} = \min_{1 \leq i \leq n} X_i$. For $k \in \{1, \dots, n\}$, let $f_k : \mathbb{R}^n \rightarrow \mathbb{R}$ denote the map $x \mapsto x_{(k)}$, so that $X_{(k)} = f_k(X_1, \dots, X_n)$. The Lipschitz constant of f_k is controlled by the following lemma.

Lemma 1.7.5. For all $x, y \in \mathbb{R}^n$ and $k \in \{1, \dots, n\}$,

$$|x_{(k)} - y_{(k)}| \leq \|x - y\|_\infty.$$

The proof of Lemma 1.7.5 is left as exercise. Since $\|x - y\|_\infty = \max_{1 \leq i \leq n} |x_i - y_i| \leq \|x - y\|_2$, the lemma gives

$$|f_k(x) - f_k(y)| \leq \|x - y\|_2 \quad \text{for all } x, y \in \mathbb{R}^n,$$

so each f_k is 1-Lipschitz. Consequently, if $(X_1, \dots, X_n) \sim N(0, I_n)$, Theorem 1.7.2 with $L = 1$ yields, for every $k = 1, \dots, n$,

$$\mathbb{P}(|X_{(k)} - \mathbb{E}[X_{(k)}]| \geq t) \leq 2e^{-\frac{t^2}{2}} \quad \text{for all } t \geq 0. \quad (1.67)$$

In particular, the fluctuations of the maximum $X_{(n)}$ around its mean are of order one, uniformly in n .

Example 1.7.6 (Gaussian complexity, [1, Example 2.30]). Let $A \subseteq \mathbb{R}^n$ be a bounded set and $W \sim N(0, I_n)$, and define

$$Z(A) := \sup_{a \in A} \langle a, W \rangle.$$

The expectation $\mathbb{E}[Z(A)]$ is called the *Gaussian complexity* of A . It is the Gaussian counterpart of the Rademacher complexity from Lecture 6, with the Rademacher vector replaced by W .

To show that $Z(A)$ concentrates, write $Z(A) = f(W)$ with $f(w) := \sup_{a \in A} \langle a, w \rangle$ for $w \in \mathbb{R}^n$, and let $D(A) := \sup_{a \in A} \|a\|_2 < \infty$. For any $w, w' \in \mathbb{R}^n$,

$$\begin{aligned} |f(w) - f(w')| &= \left| \sup_{a \in A} \langle a, w \rangle - \sup_{a \in A} \langle a, w' \rangle \right| \leq \sup_{a \in A} |\langle a, w \rangle - \langle a, w' \rangle| \\ &= \sup_{a \in A} |\langle a, w - w' \rangle| \leq \sup_{a \in A} \|a\|_2 \|w - w'\|_2 = D(A) \|w - w'\|_2. \end{aligned}$$

The second inequality is the Cauchy–Schwarz inequality. Hence f is $D(A)$ -Lipschitz, and Theorem 1.7.2 yields

$$\mathbb{P}(|Z(A) - \mathbb{E}[Z(A)]| \geq t) \leq 2 \exp\left(-\frac{t^2}{2D(A)^2}\right) \quad \text{for all } t \geq 0. \quad (1.68)$$

Example 1.7.7 (Singular values of Gaussian random matrices, [1, Example 2.32]). Let $n > d$ and let $X \in \mathbb{R}^{n \times d}$ be a random matrix with i.i.d. $N(0, 1)$ entries and ordered singular values $\sigma_1(X) \geq \sigma_2(X) \geq \dots \geq \sigma_d(X) \geq 0$. Writing $x_1^\top, \dots, x_n^\top$ for the rows of X , we view the matrix as the vector

$$\text{vec}(X) := (x_1^\top, x_2^\top, \dots, x_n^\top)^\top \in \mathbb{R}^{nd},$$

so that $\text{vec}(X) \sim N(0, I_{nd})$ and each $\sigma_k(X)$ is a function of a standard Gaussian vector in $\mathbb{R}^{nd}$. For $X, Y \in \mathbb{R}^{n \times d}$ we have $\|\text{vec}(X) - \text{vec}(Y)\|_2 = \|X - Y\|_F$, so it suffices to bound $|\sigma_k(X) - \sigma_k(Y)|$ by $\|X - Y\|_F$.

Recall that for $M \in \mathbb{R}^{n \times d}$ the spectral norm and the Frobenius norm are

$$\|M\|_2 := \sigma_1(M) = \sup_{a \in \mathbb{R}^d: \|a\|_2=1} \|Ma\|_2, \quad \|M\|_F := \left(\sum_{i=1}^n \sum_{j=1}^d M_{ij}^2 \right)^{1/2} = (\sigma_1^2(M) + \dots + \sigma_d^2(M))^{1/2},$$

so that $\|M\|_2 \leq \|M\|_F$. By Weyl's inequality (see [1, Exercise 8.3]),

$$\max_{1 \leq k \leq d} |\sigma_k(X) - \sigma_k(Y)| \leq \|X - Y\|_2 \leq \|X - Y\|_F. \quad (1.69)$$

Thus each σ_k is 1-Lipschitz as a function of $\text{vec}(X)$, and Theorem 1.7.2 gives, for each $k = 1, \dots, d$,

$$\mathbb{P}(|\sigma_k(X) - \mathbb{E}[\sigma_k(X)]| \geq t) \leq 2e^{-\frac{t^2}{2}} \quad \text{for all } t \geq 0. \quad (1.70)$$

We have not computed $\mathbb{E}[\sigma_k(X)]$, but (1.70) shows that each singular value stays within $O(1)$ of its expectation with high probability, regardless of n and d .

Chapter 2

Concentration of measure

2.1 Concentration by Entropic Techniques

We now introduce the entropy method for deriving concentration inequalities [1, Section 3.1].

Definition 2.1.1 (ϕ -entropy). Let $X \sim P$, and let ϕ be a convex function on an interval containing the support of X such that X and $\phi(X)$ are integrable. The ϕ -entropy of X is

$$H_\phi(X) := \mathbb{E}[\phi(X)] - \phi(\mathbb{E}[X]). \quad (2.1)$$

Remark 2.1.2. By Jensen's inequality, $H_\phi(X) \geq 0$. Moreover, $H_\phi(X) = 0$ if X is almost surely constant or if ϕ is affine. Although the notation suggests a function of X , $H_\phi(X)$ is a number that depends only on the distribution P of X ; for instance, $H_\phi(e^{\lambda X})$ below is the ϕ -entropy of the distribution of $e^{\lambda X}$.

Example 2.1.3. Two familiar quantities are ϕ -entropies.

- (a) If $\phi(u) = u^2$, then $H_\phi(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \text{Var}(X)$.
- (b) If $\phi(u) = -\log u$ for $u > 0$, then applying ϕ to the positive random variable $e^{\lambda X}$ gives

$$\begin{aligned} H_\phi(e^{\lambda X}) &= \mathbb{E}[-\log e^{\lambda X}] - (-\log \mathbb{E}[e^{\lambda X}]) = -\lambda \mathbb{E}[X] + \log \mathbb{E}[e^{\lambda X}] \\ &= \log e^{-\lambda \mathbb{E}[X]} + \log \mathbb{E}[e^{\lambda X}] = \log \mathbb{E}[e^{\lambda(X - \mathbb{E}[X])}], \end{aligned}$$

which is the cumulant generating function of the centered variable $X - \mathbb{E}[X]$.

From now on we work with the convex function $\phi : [0, \infty) \rightarrow \mathbb{R}$ defined by

$$\phi(u) := u \log u \quad \text{for } u > 0, \quad \phi(0) := 0, \quad (2.2)$$

where the value at 0 makes ϕ continuous, since $u \log u \rightarrow 0$ as $u \rightarrow 0^+$. For a non-negative random variable Z we drop the subscript and write

$$H(Z) := H_\phi(Z) = \mathbb{E}[Z \log Z] - \mathbb{E}[Z] \log \mathbb{E}[Z]. \quad (2.3)$$

Let $\varphi_X(\lambda) := \mathbb{E}[e^{\lambda X}]$ be the moment generating function of X (not to be confused with the convex function ϕ), and assume that it is finite in a neighborhood of λ , so that $\varphi'_X(\lambda) = \mathbb{E}[Xe^{\lambda X}]$. Taking $Z = e^{\lambda X}$ in (2.3) and using $\log e^{\lambda X} = \lambda X$, we obtain

$$H(e^{\lambda X}) = \mathbb{E}[e^{\lambda X} \cdot \lambda X] - \mathbb{E}[e^{\lambda X}] \log \mathbb{E}[e^{\lambda X}] = \lambda \varphi'_X(\lambda) - \varphi_X(\lambda) \log \varphi_X(\lambda). \quad (2.4)$$

Hence $H(e^{\lambda X})$ is determined by the moment generating function of X and its derivative.

Example 2.1.4 (Entropy of a Gaussian variable, [1, Example 3.1]). Let $X \sim N(0, \sigma^2)$. Then $\varphi_X(\lambda) = e^{\lambda^2 \sigma^2 / 2}$ and $\varphi'_X(\lambda) = \lambda \sigma^2 e^{\lambda^2 \sigma^2 / 2}$ for all $\lambda \in \mathbb{R}$, so by (2.4),

$$H(e^{\lambda X}) = \lambda \cdot \lambda \sigma^2 e^{\frac{\lambda^2 \sigma^2}{2}} - e^{\frac{\lambda^2 \sigma^2}{2}} \log e^{\frac{\lambda^2 \sigma^2}{2}} = \frac{\lambda^2 \sigma^2}{2} e^{\frac{\lambda^2 \sigma^2}{2}} = \frac{1}{2} \sigma^2 \lambda^2 \varphi_X(\lambda). \quad (2.5)$$

Since the moment generating function yields tail bounds through the Chernoff bound, (2.4) suggests that an upper bound on $H(e^{\lambda X})$ should also lead to a tail bound. The Herbst argument makes this precise for upper bounds of the Gaussian form (2.5).

Proposition 2.1.5 (Herbst argument). *Let $\sigma > 0$ be a constant, $\varphi_X(\lambda) = \mathbb{E}[e^{\lambda X}]$, and $I = \mathbb{R}$ or $I = [0, +\infty)$. Suppose the entropy $H(e^{\lambda X})$ satisfies*

$$H(e^{\lambda X}) \leq \frac{1}{2} \sigma^2 \lambda^2 \varphi_X(\lambda) \quad \text{for all } \lambda \in I. \quad (2.6)$$

Then X satisfies

$$\log \mathbb{E}[e^{\lambda(X - \mathbb{E}[X])}] \leq \frac{1}{2} \lambda^2 \sigma^2 \quad \text{for all } \lambda \in I, \quad (2.7)$$

or equivalently, $\mathbb{E}[e^{\lambda(X - \mathbb{E}[X])}] \leq e^{\frac{\lambda^2 \sigma^2}{2}}$ for all $\lambda \in I$.

Proof of Proposition 2.1.5. Case 1: $I = [0, \infty)$.

① **A differential inequality for the MGF.** Combining (2.4) with the hypothesis (2.6) gives

$$\lambda \varphi'_X(\lambda) - \varphi_X(\lambda) \log \varphi_X(\lambda) \leq \frac{1}{2} \sigma^2 \lambda^2 \varphi_X(\lambda) \quad \text{for all } \lambda \in I. \quad (2.8)$$

Since $\varphi_X(\lambda) > 0$, we can divide by $\varphi_X(\lambda)$:

$$\frac{\lambda \varphi'_X(\lambda)}{\varphi_X(\lambda)} - \log \varphi_X(\lambda) \leq \frac{1}{2} \sigma^2 \lambda^2, \quad \lambda \geq 0. \quad (2.9)$$

② **The auxiliary function G .** The key step is to rewrite the left side of (2.9) as a derivative. Define

$$G(\lambda) := \frac{\log \varphi_X(\lambda)}{\lambda}, \quad \lambda > 0.$$

By the product rule,

$$G'(\lambda) = -\frac{1}{\lambda^2} \log \varphi_X(\lambda) + \frac{1}{\lambda} \cdot \frac{\varphi'_X(\lambda)}{\varphi_X(\lambda)} = \frac{1}{\lambda^2} \left[\frac{\lambda \varphi'_X(\lambda)}{\varphi_X(\lambda)} - \log \varphi_X(\lambda) \right].$$

So, up to the factor $1/\lambda^2$, $G'(\lambda)$ is exactly the left side of (2.9). Dividing (2.9) by $\lambda^2 > 0$ gives

$$G'(\lambda) \leq \frac{1}{2} \sigma^2 \quad \text{for all } \lambda > 0. \quad (2.10)$$

③ **Integrate.** Fix $\lambda > 0$. For any $\lambda_0 \in (0, \lambda]$, by the fundamental theorem of calculus and (2.10),

$$G(\lambda) - G(\lambda_0) = \int_{\lambda_0}^{\lambda} G'(s) ds \leq \int_{\lambda_0}^{\lambda} \frac{1}{2} \sigma^2 ds = \frac{1}{2} \sigma^2 (\lambda - \lambda_0).$$

(We need $\lambda_0 \leq \lambda$ here: if $\lambda < \lambda_0$, the integral runs backwards and the inequality flips. This does no harm, because we next send $\lambda_0 \rightarrow 0^+$.)

④ **Let** $\lambda_0 \rightarrow 0^+$. Taking the limit (which exists, as computed below),

$$G(\lambda) - \lim_{\lambda_0 \rightarrow 0^+} G(\lambda_0) \leq \frac{1}{2}\sigma^2\lambda.$$

To compute the limit, use $\varphi_X(0) = \mathbb{E}[e^0] = 1$, so $\log \varphi_X(0) = 0$. The limit is then a right-hand difference quotient:

$$\begin{aligned} \lim_{\lambda_0 \rightarrow 0^+} G(\lambda_0) &= \lim_{\lambda_0 \rightarrow 0^+} \frac{\log \varphi_X(\lambda_0)}{\lambda_0} = \lim_{\lambda_0 \rightarrow 0^+} \frac{\log \varphi_X(\lambda_0) - \log \varphi_X(0)}{\lambda_0 - 0} = \frac{d}{d\lambda} \log \varphi_X(\lambda) \Big|_{\lambda=0} \\ &= \frac{\varphi'_X(0)}{\varphi_X(0)} = \frac{\mathbb{E}[X]}{1} = \mathbb{E}[X]. \end{aligned}$$

So G extends continuously to 0 with $G(0) := \mathbb{E}[X]$.

⑤ **Unwind.** We have shown that for all $\lambda > 0$,

$$G(\lambda) - \mathbb{E}[X] \leq \frac{1}{2}\sigma^2\lambda \iff \frac{\log \varphi_X(\lambda)}{\lambda} - \mathbb{E}[X] \leq \frac{1}{2}\sigma^2 \iff \log \varphi_X(\lambda) - \lambda \mathbb{E}[X] \leq \frac{1}{2}\sigma^2\lambda^2.$$

Since $\log \varphi_X(\lambda) - \lambda \mathbb{E}[X] = \log \mathbb{E}[e^{\lambda X}] - \log e^{\lambda \mathbb{E}[X]} = \log \mathbb{E}[e^{\lambda(X - \mathbb{E}[X])}]$, this is

$$\log \mathbb{E}[e^{\lambda(X - \mathbb{E}[X])}] \leq \frac{1}{2}\sigma^2\lambda^2 \quad \text{for all } \lambda > 0.$$

The bound also holds at $\lambda = 0$, where both sides equal 0. This proves the claim for $I = [0, \infty)$.

Case 2: $I = \mathbb{R}$. The text leaves this case as an exercise. For $\lambda < 0$, (2.8) still holds. Dividing by $\lambda^2 \varphi_X(\lambda) > 0$ again gives $G'(\lambda) \leq \frac{1}{2}\sigma^2$ for all $\lambda < 0$. Fix $\lambda < \lambda_0 < 0$ and integrate:

$$G(\lambda_0) - G(\lambda) \leq \frac{1}{2}\sigma^2(\lambda_0 - \lambda).$$

Let $\lambda_0 \rightarrow 0^-$. The same difference-quotient argument gives $G(\lambda_0) \rightarrow \mathbb{E}[X]$, so $\mathbb{E}[X] - G(\lambda) \leq -\frac{1}{2}\sigma^2\lambda$, that is, $G(\lambda) \geq \mathbb{E}[X] + \frac{1}{2}\sigma^2\lambda$. Multiplying by $\lambda < 0$ *reverses* the inequality:

$$\log \varphi_X(\lambda) = \lambda G(\lambda) \leq \lambda \mathbb{E}[X] + \frac{1}{2}\sigma^2\lambda^2,$$

which is (2.7) for $\lambda < 0$. □

Remark 2.1.6 (Why G ?). The function $\lambda G(\lambda) = \log \varphi_X(\lambda)$ is the cumulant generating function. The target (2.7) reads $\log \varphi_X(\lambda) \leq \lambda \mathbb{E}[X] + \frac{1}{2}\sigma^2\lambda^2$, which is $G(\lambda) \leq \mathbb{E}[X] + \frac{1}{2}\sigma^2\lambda$. So it is enough to show that G starts at $\mathbb{E}[X]$ and has slope at most $\sigma^2/2$. This reduces the problem to a one-line integration.

Remark 2.1.7 (Interpretation and consequences).

(a) If $I = \mathbb{R}$, then (2.7) says exactly that $X - \mathbb{E}[X]$ is *sub-Gaussian with parameter* σ .

(b) If $I = [0, \infty)$, the Chernoff method still gives the one-sided bound: for every $t \geq 0$,

$$\mathbb{P}[X \geq \mathbb{E}[X] + t] \leq \inf_{\lambda \geq 0} e^{-\lambda t} \mathbb{E}[e^{\lambda(X - \mathbb{E}[X])}] \leq \inf_{\lambda \geq 0} e^{-\lambda t + \frac{\lambda^2 \sigma^2}{2}} = e^{-\frac{t^2}{2\sigma^2}},$$

where the infimum is attained at $\lambda^* = t/\sigma^2$.

In short, *entropy control* $\Rightarrow$ *MGF control* $\Rightarrow$ *tail control*. The entropy method is useful because entropy *tensorizes* over independent coordinates, which the MGF does not do as easily.

2.2 Wasserstein Distances

This section follows Section 3.3.1 of [1]. The material leads into optimal transport and transportation cost inequalities.

Definition 2.2.1 (Lipschitz functions). Let $(\mathcal{X}, \rho)$ be a metric space. A function $f : \mathcal{X} \rightarrow \mathbb{R}$ is *L-Lipschitz with respect to ρ* if

$$|f(x) - f(x')| \leq L \cdot \rho(x, x') \quad \text{for all } x, x' \in \mathcal{X}.$$

We write $\|f\|_{\text{Lip}}$ for the smallest L for which this holds:

$$\|f\|_{\text{Lip}} = \sup_{x \neq x'} \frac{|f(x) - f(x')|}{\rho(x, x')}.$$

Definition 2.2.2 (Wasserstein distance induced by ρ). Given two probability distributions Q, P on $\mathcal{X}$, define

$$W_\rho(Q, P) := \sup_{\|f\|_{\text{Lip}} \leq 1} \left(\int f dQ - \int f dP \right). \quad (2.11)$$

This is the *Wasserstein distance induced by ρ* (also called the W_1 or Kantorovich–Rubinstein distance).

Remark 2.2.3 (Invariance to constant offsets). For any constant c , $\int (f + c) dQ - \int (f + c) dP = \int f dQ - \int f dP$, because both measures have total mass 1. So in (2.13) we can shift each f by any constant without changing the value.

2.3 Example: Hamming Metric and Total Variation

Consider the *Hamming (discrete) metric*

$$\rho(x, x') = \mathbf{1}_{\{x \neq x'\}}.$$

Throughout, all sets and functions are measurable.

Definition 2.3.1 (Three definitions of total variation). For probability distributions Q, P on $\mathcal{X}$:

- (a) (Sets) $\|Q - P\|_{\text{TV}} = \sup_{A \subseteq \mathcal{X}} |Q(A) - P(A)|$.
- (b) (Test functions) $\|Q - P\|_{\text{TV}} = \sup_{f: \mathcal{X} \rightarrow [0, 1]} \left(\int f dQ - \int f dP \right)$.
- (c) (Densities) If P and Q have densities p and q with respect to a base (dominating) measure ν ,

$$\|Q - P\|_{\text{TV}} = \frac{1}{2} \int |p(x) - q(x)| \nu(dx).$$

Remark 2.3.2. A dominating measure always exists: take $\nu = \frac{1}{2}(P + Q)$, so that both $P, Q \ll \nu$. The value in Definition 2.3.1(c) does not depend on which dominating ν is chosen.

Proposition 2.3.3. *For the Hamming metric, W_ρ equals the total variation distance:*

$$W_{\text{Ham}}(Q, P) = \|Q - P\|_{\text{TV}} := \sup_{A \subseteq \mathcal{X}} |Q(A) - P(A)|,$$

where the supremum ranges over all measurable A .

Proof of Proposition 2.3.3. Key observation. If $\|f\|_{\text{Lip}} \leq 1$ with respect to the Hamming metric, then $|f(x) - f(x')| \leq 1 \cdot \rho(x, x') \leq 1$ for all x, x' . Conversely, *any* $f : \mathcal{X} \rightarrow [0, 1]$ satisfies $|f(x) - f(x')| \leq \mathbf{1}_{\{x \neq x'\}}$, so it is 1-Lipschitz.

Reduction to $[0, 1]$ -valued functions. By Remark 2.2.3, W_ρ does not change if f is shifted by a constant. If $\|f\|_{\text{Lip}} \leq 1$, the key observation gives $\sup f - \inf f \leq 1$, so $\tilde{f} := f - \inf f$ takes values in $[0, 1]$ and gives the same objective. Hence we may restrict the supremum to functions with $f(x) \in [0, 1]$ for all $x \in \mathcal{X}$:

$$\begin{aligned} W_{\text{Ham}}(Q, P) &= \sup_{\substack{\|f\|_{\text{Lip}} \leq 1 \\ f: \mathcal{X} \rightarrow [0, 1]}} \left(\int f dQ - \int f dP \right) \\ &= \sup_{f: \mathcal{X} \rightarrow [0, 1]} \left(\int f dQ - \int f dP \right) && \text{(key observation)} \\ &= \|Q - P\|_{\text{TV}} && ([1, \text{Exercise 3.13}]). \end{aligned}$$

The second line is exactly Definition 2.3.1(b), and the last equality is the equivalence of Definitions 2.3.1(a) and 2.3.1(b) (homework #2). $\square$

Remark 2.3.4. Putting everything together, for the Hamming metric

$$W_{\text{Ham}}(Q, P) = \sup_{f: \mathcal{X} \rightarrow [0, 1]} \int f (dQ - dP) = \|Q - P\|_{\text{TV}} = \frac{1}{2} \int |p - q| d\nu = \inf_M M[X \neq X'].$$

Here the equality with $\frac{1}{2} \int |p - q| d\nu$ is Definition 2.3.1(c), and the coupling form $\inf_M M[X \neq X']$ is also part of [1, Exercise 3.13]. In words, W_{Ham} is one half of the $L^1(\nu)$ distance between the densities. The last equality is a concrete instance of the Kantorovich–Rubinstein duality in the next section.

2.4 Optimal Transport and Couplings

Definition 2.4.1 (Coupling). A distribution M on $\mathcal{X} \otimes \mathcal{X}$ is a *coupling* of the pair (Q, P) if its marginal distributions in the first and second coordinates are Q and P respectively:

$$M(A \times \mathcal{X}) = Q(A), \quad M(\mathcal{X} \times B) = P(B) \quad \text{for all } A, B.$$

Equivalently, $(X, X') \sim M$ with $X \sim Q$ and $X' \sim P$. Couplings always exist; for example, the product $M = Q \otimes P$.

Transport cost. The quantity

$$\inf_{\substack{M \text{ a coupling} \\ \text{of } (Q, P)}} \mathbb{E}_M [\rho(X, X')], \quad (X, X') \sim M,$$

is also a distance between Q and P . It is the minimal expected cost of moving mass from Q to P when moving a unit of mass from x to x' costs $\rho(x, x')$.

2.4.1 Weak Duality

Proposition 2.4.2 (Weak duality: $\inf \geq \sup$). *Assume Q and P have finite first moments. Then*

$$\inf_{\substack{M \text{ coupling} \\ \text{of } (Q, P)}} \int \rho(x, x') dM(x, x') \geq \sup_{\|f\|_{\text{Lip}} \leq 1} \left(\int f dQ - \int f dP \right) = W_\rho(Q, P). \quad (2.12)$$

Proof. Every 1-Lipschitz f is integrable, since $|f(x)| \leq |f(x_0)| + \rho(x, x_0)$. Fix any f with $\|f\|_{\text{Lip}} \leq 1$ and any coupling M of (Q, P) . Then

$$\begin{aligned}
\int \rho(x, x') dM(x, x') &\geq \int (f(x) - f(x')) dM(x, x') && (\text{since } \|f\|_{\text{Lip}} \leq 1) \\
&= \int f(x) dM(x, x') - \int f(x') dM(x, x') && (\text{linearity}) \\
&= \int f(x) dQ(x) - \int f(x') dP(x') && (\text{marginals of } M \text{ are } Q, P) \\
&= \int f dQ - \int f dP.
\end{aligned}$$

The left side does not depend on f and the right side does not depend on M . Taking the infimum over couplings on the left and the supremum over 1-Lipschitz f on the right gives (2.12). Kantorovich–Rubinstein duality says equality always holds $\square$

We begin by completing the discussion of distances and divergences between probability measures. Let $(\mathcal{X}, \rho)$ be a metric space, and let $\mathbb{Q}$ and $\mathbb{P}$ be probability distributions on $\mathcal{X}$.

2.4.2 Optimal Transport Theory

Optimal transport theory studies how to transform one probability distribution $\mathbb{Q}$ into another probability distribution $\mathbb{P}$ while minimizing the cost of transporting probability mass.

A *transportation plan*, or *coupling*, of $\mathbb{Q}$ and $\mathbb{P}$ is a joint probability distribution M of a pair (X, X') such that

$$X \sim \mathbb{Q} \quad \text{and} \quad X' \sim \mathbb{P}.$$

The collection of all couplings of $\mathbb{Q}$ and $\mathbb{P}$ is denoted by $\Pi(\mathbb{Q}, \mathbb{P})$.

Let $\rho(x, x')$ denote the cost of moving one unit of mass from x to x' . For a coupling $M \in \Pi(\mathbb{Q}, \mathbb{P})$, its expected transportation cost is

$$\mathbb{E}_M [\rho(X, X')].$$

Definition 2.4.3 (Wasserstein distance). For $s \geq 1$, the order- s Wasserstein distance is

$$W_{\rho, s}(\mathbb{Q}, \mathbb{P}) := \inf_{M \in \Pi(\mathbb{Q}, \mathbb{P})} (\mathbb{E}_M [\rho(X, X')^s])^{1/s}. \quad (2.13)$$

When $s = 1$, we write $W_{\rho}(\mathbb{Q}, \mathbb{P}) := W_{\rho, 1}(\mathbb{Q}, \mathbb{P})$. The case $s = 2$ is also frequently used in practice.

Example 2.4.4 (Transporting bread to coffee stores). Suppose that bread is produced at two store locations, represented by -1 and 1 . The probability measure

$$\mathbb{Q} = \frac{1}{2}\delta_{-1} + \frac{1}{2}\delta_1$$

describes the distribution of the bread supply. Thus, one half of the bread is produced at location -1 , and the other half is produced at location 1 .

The bread must be delivered to coffee stores at the same two locations. The probability measure

$$\mathbb{P} = \frac{1}{3}\delta_{-1} + \frac{2}{3}\delta_1$$

describes the distribution of demand. The coffee store at location -1 requires one third of the total bread, whereas the coffee store at location 1 requires two thirds.

Let $m(x, x')$ denote the amount of bread transported from production location x under $\mathbb{Q}$ to coffee-store location x' under $\mathbb{P}$. The four possible transportation quantities are

$$m(-1, -1), \quad m(-1, 1), \quad m(1, -1), \quad m(1, 1).$$

They have the following interpretations:

- $m(-1, -1)$ is the amount transported from the bread store at -1 to the coffee store at -1 ;
- $m(-1, 1)$ is the amount transported from the bread store at -1 to the coffee store at 1 ;
- $m(1, -1)$ is the amount transported from the bread store at 1 to the coffee store at -1 ;
- $m(1, 1)$ is the amount transported from the bread store at 1 to the coffee store at 1 .

The supply and demand constraints are

$$\begin{aligned} m(-1, -1) + m(-1, 1) &= \frac{1}{2}, & m(1, -1) + m(1, 1) &= \frac{1}{2}, \\ m(-1, -1) + m(1, -1) &= \frac{1}{3}, & m(-1, 1) + m(1, 1) &= \frac{2}{3}. \end{aligned}$$

An optimal transportation plan is

$$\begin{aligned} m(-1, -1) &= \frac{1}{3}, & m(-1, 1) &= \frac{1}{6}, \\ m(1, -1) &= 0, & m(1, 1) &= \frac{1}{2}. \end{aligned}$$

Equivalently, the plan is represented by the table

	Coffee store -1	Coffee store 1	Bread supply
Bread store -1	$m(-1, -1) = \frac{1}{3}$	$m(-1, 1) = \frac{1}{6}$	$\frac{1}{2}$
Bread store 1	$m(1, -1) = 0$	$m(1, 1) = \frac{1}{2}$	$\frac{1}{2}$
Coffee-store demand	$\frac{1}{3}$	$\frac{2}{3}$	1

Suppose that $\rho(x, x') = |x - x'|$. Transporting bread between stores at the same location has zero cost, whereas transporting bread from one location to the other has cost 2. Therefore, the total cost is

$$\begin{aligned} &\rho(-1, -1)m(-1, -1) + \rho(-1, 1)m(-1, 1) \\ &\quad + \rho(1, -1)m(1, -1) + \rho(1, 1)m(1, 1) \\ &= 0 \left(\frac{1}{3} \right) + 2 \left(\frac{1}{6} \right) + 2(0) + 0 \left(\frac{1}{2} \right) = \frac{1}{3}. \end{aligned}$$

The marginal constraints force at least $1/6$ of the bread produced at -1 to be transported to 1 . Hence no coupling can have a smaller cost, and

$$W_\rho(\mathbb{Q}, \mathbb{P}) = \frac{1}{3}.$$

Suppose Q and P have densities q and p on $\mathcal{X}$, and a coupling M has density $m(x, x')$. Think of q as an initial pile of mass and p as the target configuration. Then:

- $\rho(x, x')$ is the cost per unit of moving mass from x to x' ;
- $m(x, x') dx dx'$ is the amount of mass sent from near x to near x' (a *transport plan*);
- the marginal constraints say the plan removes exactly q and delivers exactly p ;

- the total cost is $\iint \rho(x, x') m(x, x') dx dx'$.

Minimizing over admissible plans gives the Wasserstein distance. This is why it is called the “earth mover’s” or transportation distance.

For any 1-Lipschitz function f and any coupling $M \in \Pi(\mathbb{Q}, \mathbb{P})$,

$$\begin{aligned} \int_{\mathcal{X}} f d\mathbb{Q} - \int_{\mathcal{X}} f d\mathbb{P} &= \mathbb{E}_M [f(X) - f(X')] \\ &\leq \mathbb{E}_M [\rho(X, X')]. \end{aligned}$$

Taking the supremum over all 1-Lipschitz functions and then the infimum over all couplings gives the weak-duality inequality.

Theorem 2.4.5 (Kantorovich–Rubinstein duality). *Under the usual regularity and finite-first-moment assumptions,*

$$\begin{aligned} W_\rho(\mathbb{Q}, \mathbb{P}) &= \sup_{\|f\|_{\text{Lip}} \leq 1} \left[\int_{\mathcal{X}} f d\mathbb{Q} - \int_{\mathcal{X}} f d\mathbb{P} \right] \\ &= \inf_{M \in \Pi(\mathbb{Q}, \mathbb{P})} \int_{\mathcal{X} \times \mathcal{X}} \rho(x, x') dM(x, x') \\ &= \inf_{M \in \Pi(\mathbb{Q}, \mathbb{P})} \mathbb{E}_M [\rho(X, X')]. \end{aligned} \tag{2.14}$$

If $\mathbb{Q}$ and $\mathbb{P}$ have densities q and p , and the coupling M has density m , then the transportation cost associated with m is

$$\int_{\mathcal{X} \times \mathcal{X}} \rho(x, x') m(x, x') dx dx'.$$

Thus, the primal formulation (coupling formulation) minimizes transportation cost, whereas the dual formulation (Lipschitz function formulation) compares expectations over 1-Lipschitz functions.

2.4.3 Kullback–Leibler Divergence

Definition 2.4.6 (Kullback–Leibler divergence). The Kullback–Leibler (KL) divergence of $\mathbb{Q}$ from $\mathbb{P}$ is

$$D(\mathbb{Q} \parallel \mathbb{P}) := \begin{cases} \mathbb{E}_{\mathbb{Q}} \left[\log \frac{d\mathbb{Q}}{d\mathbb{P}} \right], & \mathbb{Q} \ll \mathbb{P}, \\ +\infty, & \text{otherwise.} \end{cases} \tag{2.15}$$

If $\mathbb{Q}$ and $\mathbb{P}$ have densities q and p with respect to a common dominating measure ν , then

$$D(\mathbb{Q} \parallel \mathbb{P}) = \int_{\mathcal{X}} q(x) \log \frac{q(x)}{p(x)} \nu(dx). \tag{2.16}$$

Unlike a metric, KL divergence is generally asymmetric and does not satisfy the triangle inequality. It nevertheless controls total variation distance.

Theorem 2.4.7 (Pinsker–Csiszár–Kullback inequality). *With the convention*

$$\|\mathbb{P} - \mathbb{Q}\|_{\text{TV}} := \sup_{A \subseteq \mathcal{X}} |\mathbb{P}(A) - \mathbb{Q}(A)|,$$

we have

$$\|\mathbb{P} - \mathbb{Q}\|_{\text{TV}} \leq \sqrt{\frac{1}{2} D(\mathbb{Q} \parallel \mathbb{P})}. \tag{2.17}$$

Thus, small KL divergence implies that the two measures assign similar probabilities to every measurable event.

Chapter 3

Uniform Laws of Large Numbers

3.1 Motivation

3.1.1 Uniform Convergence of CDFs

Let X be a real-valued random variable with cumulative distribution function

$$F(t) := \mathbb{P}(X \leq t) = \mathbb{E} [\mathbf{1}_{\{X \leq t\}}], \quad t \in \mathbb{R}.$$

Suppose that

$$X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Law}(X).$$

Definition 3.1.1 (Empirical CDF). The empirical cumulative distribution function is

$$\hat{F}_n(t) := \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{\{X_i \leq t\}}, \quad t \in \mathbb{R}. \quad (3.1)$$

For every fixed t , the strong law of large numbers gives

$$\hat{F}_n(t) \longrightarrow F(t) \quad \text{almost surely.}$$

This is pointwise convergence: the exceptional event may initially depend on t .

Many estimation problems can be formulated through a functional γ that maps a CDF F to a real number $\gamma(F)$. The plug-in estimator is $\gamma(\hat{F}_n)$, and the basic consistency question is whether

$$\gamma(\hat{F}_n) \xrightarrow{?} \gamma(F).$$

Example 3.1.2 (Expectation functionals). Let g be an integrable function and define

$$\gamma_g(F) := \int_{\mathbb{R}} g(x) dF(x).$$

Because $\hat{F}_n$ assigns mass $1/n$ to each observation,

$$\gamma_g(\hat{F}_n) = \int_{\mathbb{R}} g(x) d\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n g(X_i).$$

Therefore, if $\mathbb{E} |g(X)| < \infty$, the strong law gives

$$\gamma_g(\hat{F}_n) \longrightarrow \gamma_g(F) \quad \text{almost surely.}$$

Example 3.1.3 (Quantile functionals). For $\alpha \in (0, 1)$, the lower α -quantile functional is

$$Q_\alpha(F) := \inf \{t \in \mathbb{R} : F(t) \geq \alpha\}.$$

The corresponding empirical quantile is

$$\begin{aligned} Q_\alpha(\hat{F}_n) &= \inf \left\{ t \in \mathbb{R} : \hat{F}_n(t) \geq \alpha \right\} \\ &= \inf \left\{ t \in \mathbb{R} : \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{\{X_i \leq t\}} \geq \alpha \right\}. \end{aligned}$$

It is not immediately clear from pointwise convergence alone whether $Q_\alpha(\hat{F}_n) \rightarrow Q_\alpha(F)$. Uniform convergence and an appropriate condition ensuring that F crosses the level α are needed.

Definition 3.1.4 (Supremum norm and continuity). For two CDFs F and G , their supremum distance is

$$\|F - G\|_\infty := \sup_{t \in \mathbb{R}} |F(t) - G(t)|. \quad (3.2)$$

A functional γ is continuous at F in the supremum norm if, for every $\epsilon > 0$, there exists $\delta > 0$ such that

$$\|G - F\|_\infty \leq \delta \implies |\gamma(G) - \gamma(F)| \leq \epsilon.$$

Recall that if f is a continuous function on $\mathbb{R}^d$, then

$$x_n \rightarrow x \implies f(x_n) \rightarrow f(x).$$

Similarly, if $F_n \rightarrow F$ in the supremum norm and γ is continuous at F , then

$$\gamma(F_n) \rightarrow \gamma(F).$$

Therefore, to establish the consistency of $\gamma(\hat{F}_n)$, it is sufficient to verify that

1. $\|\hat{F}_n - F\|_\infty \rightarrow 0$;
2. γ is continuous at F with respect to the supremum norm.

Theorem 3.1.5 (Glivenko–Cantelli). *For every distribution function F , the empirical CDF $\hat{F}_n$ is a strongly consistent estimator of the population CDF in the supremum norm:*

$$\|\hat{F}_n - F\|_\infty \longrightarrow 0 \quad \text{almost surely.} \quad (3.3)$$

The proof is deferred.

3.1.2 Uniform Laws for General Function Classes

The empirical-CDF result is a special case of a general problem involving empirical averages over a collection of functions. Let

$$X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \mathbb{P}$$

on a measurable space $\mathcal{X}$.

Definition 3.1.6 (Empirical measure). The empirical probability measure is

$$\mathbb{P}_n := \frac{1}{n} \sum_{i=1}^n \delta_{X_i}. \quad (3.4)$$

For $a \in \mathcal{X}$, the Dirac probability measure δ_a is defined by

$$\delta_a(B) := \begin{cases} 1, & a \in B, \\ 0, & a \notin B, \end{cases} \quad B \subseteq \mathcal{X}. \quad (3.5)$$

Example 3.1.7 (Bernoulli distribution). The Bernoulli distribution with parameter p can be written as

$$\text{Ber}(p) = p\delta_1 + (1 - p)\delta_0.$$

Its probability mass function is

$$\text{pmf}(x) = \begin{cases} p, & x = 1, \\ 1 - p, & x = 0. \end{cases}$$

Thus, if $\mathbb{P} = p\delta_1 + (1 - p)\delta_0$, then

$$\mathbb{P}(\{1\}) = p, \quad \mathbb{P}(\{0\}) = 1 - p.$$

Definition 3.1.8 (Population and empirical averages). For a measurable function f , define

$$\mathbb{P}f := \int_{\mathcal{X}} f(x) d\mathbb{P}(x) = \mathbb{E}_{\mathbb{P}}[f(X)]. \quad (3.6)$$

The corresponding empirical average is

$$\mathbb{P}_n f := \int_{\mathcal{X}} f(x) d\mathbb{P}_n(x) = \frac{1}{n} \sum_{i=1}^n f(X_i). \quad (3.7)$$

For each fixed integrable function f , the strong law of large numbers implies

$$\mathbb{P}_n f \longrightarrow \mathbb{P}f \quad \text{almost surely.}$$

Definition 3.1.9. Let $\mathcal{F}$ be a class of measurable functions. Define

$$\|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}} := \sup_{f \in \mathcal{F}} |\mathbb{P}_n f - \mathbb{P}f|. \quad (3.8)$$

The class $\mathcal{F}$ is called a $\mathbb{P}$ -Glivenko–Cantelli class if

$$\|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}} \longrightarrow 0 \quad \text{almost surely.} \quad (3.9)$$

Example 3.1.10 (Empirical CDFs and indicator functions). Define the threshold class

$$\mathcal{F}_{\text{cdf}} := \{f_t(\cdot) : t \in \mathbb{R}\}, \quad f_t(x) := \mathbf{1}_{\{x \leq t\}}.$$

For every $t \in \mathbb{R}$,

$$\mathbb{P}f_t = \int_{\mathcal{X}} \mathbf{1}_{\{x \leq t\}} d\mathbb{P}(x) = \mathbb{P}(X \leq t) = F(t),$$

and

$$\mathbb{P}_n f_t = \frac{1}{n} \sum_{i=1}^n f_t(X_i) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{\{X_i \leq t\}} = \widehat{F}_n(t).$$

Therefore,

$$\|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}_{\text{cdf}}} = \sup_{t \in \mathbb{R}} |\widehat{F}_n(t) - F(t)| = \|\widehat{F}_n - F\|_{\infty}.$$

The Glivenko–Cantelli theorem says precisely that the threshold class $\mathcal{F}_{\text{cdf}}$ is a universal Glivenko–Cantelli class.

This reformulation is the bridge from empirical CDFs to empirical-process theory. For a single function, the law of large numbers is enough. For an entire class, the size and structure of the class determine whether the same sample controls all empirical averages simultaneously.

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